

The Future of Shipping in the Arctic:

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The Future of Shipping in the Arctic: New Perspectives on the Next Frontier
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Executive Summary

This year, levels of ice in the Arctic were the seventh lowest on record.[1] As ice has receded, there has been a corresponding increase in focus on the Arctic North Atlantic as the next area of great industrial expansion. However, despite increased media attention, and a number of predictions that trans-Arctic shipping was about to explode in the region, actual levels of shipping have been muted relative to the publicity. There have been increases, however, in the size and number of ships undertaking destinational voyages along established routes. The current study represents a forecast analysis examining increased levels of shipping in the Arctic, paying specific attention to industrial development in the ANA region.

For this analysis, we adopted a three-pronged approach. First, we examined historical data on incidents in the Arctic and North Atlantic region, in order to see which characteristics defined ship casualties in the past. We then reviewed a sample of after-action reports from tabletop or live action exercises conducted in the Arctic, in effort to determine which scenarios first responders felt were most likely, or most in need of preparation. Finally, we conducted 28 interviews with a wide variety of Arctic stakeholders, including representatives from shipping companies, think tanks, maritime insurance, cruise industry, government and coastal response agencies. Our interview participants also included academics, and members of Arctic Council working groups. Compiling data from all three sources of information provided us with a rich understanding of the area under consideration, although it should be noted that we were unable to interview representatives from important stakeholder groups, which may lead to gaps in understanding.

There were a number of themes which emerged from the data analyzed in the current study, the most prominent of which include:

- Arctic shipping is all about the economics
- We might need to expand our focus
- For emergency preparedness, it is important to be proactive rather than reactive
- We need to capitalize on existing strengths
- There are very few absolutes in the Arctic

A total of 11 recommendations resulted from the current analysis, some of which are more likely to be actionable at the present time than others. The most pressing recommendations include:

1. Expansion of both tabletop and live exercises to include features of historical incidents, and increased participation from both commercial industries and local communities.
2. Explore public private partnerships in emergency management, especially as they relate to the potential inherent in vessels of opportunity.
3. Expand the current Polar Code training requirements for crew operating vessels in the Arctic, to include actual ice navigation experience, and first responder training. Also, expand required training to all crew, not just officers.
4. Open access to data regarding shipping incidents and traffic. Consider ways in which existing databases may be expanded to include relevant information needed in most emergency situations. Also consider ways to separate out classified from unclassified information to increase the effectiveness of information sharing in emergency response scenarios.

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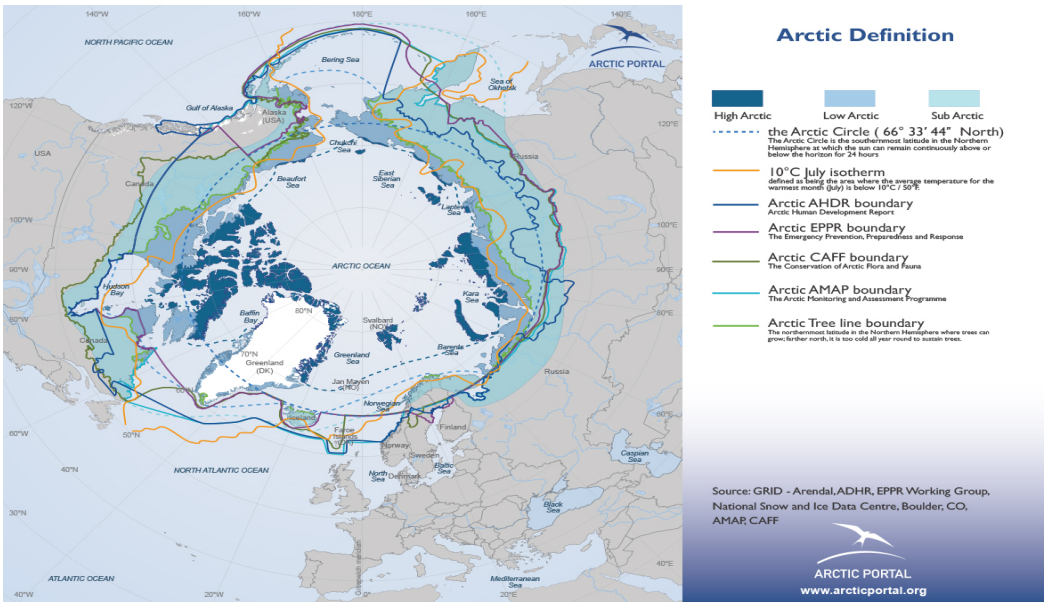
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The Future of Shipping in the Arctic:
New Perspectives on the Next Frontier

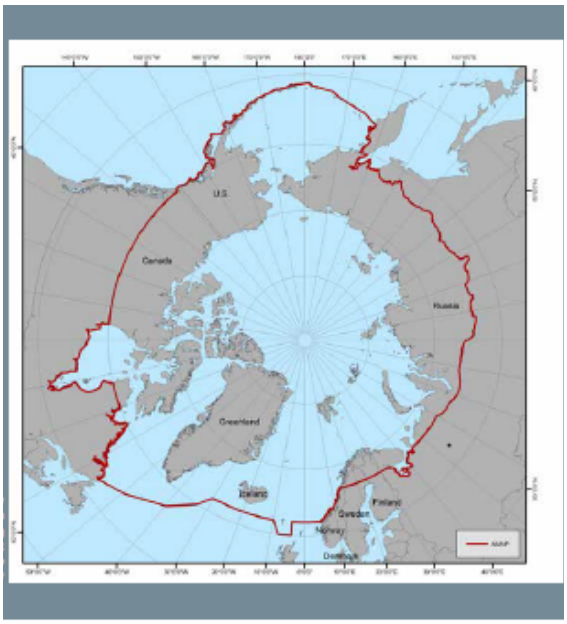
The top of the world is melting, and at a much faster rate than anyone anticipated. The winter of 2018 brought new record lows in the extent of sea ice in the Arctic, nearly 500,000 square miles less than the historic average[2]. Accompanying these changes has been an increased focus on the Arctic as the next center of global development with predicted increases in shipping, infrastructure and economic development. But while experts have been predicting a mass influx of shipping coming to the Arctic, actual development has been much more gradual. At present, shipping is constrained largely to existing industries, utilizing established routes. The goal of the current study was to prepare a foresight analysis for shipping in the Arctic, predicting trends that can be expected in the coming years, and highlighting policy gaps in the current regulatory framework.

Defining Arctic, Arctic North Atlantic

One of the early difficulties of any discussion of the Arctic comes from the fact that there are many different definitions of "the Arctic." For example, in the map below compiled by the Arctic Portal, a non-profit focused on information sharing in the region, we see seven different definitions of what constitutes the Arctic. [3]



The variety of definitions come from different countries or organizations that have different priorities in the region. Below we show some of the boundaries that distinguish the Arctic from the rest of the globe. Defining what is "Arctic" and what is "sub-Arctic" can have real implications for policy creation and can make cooperation and international relations challenging.

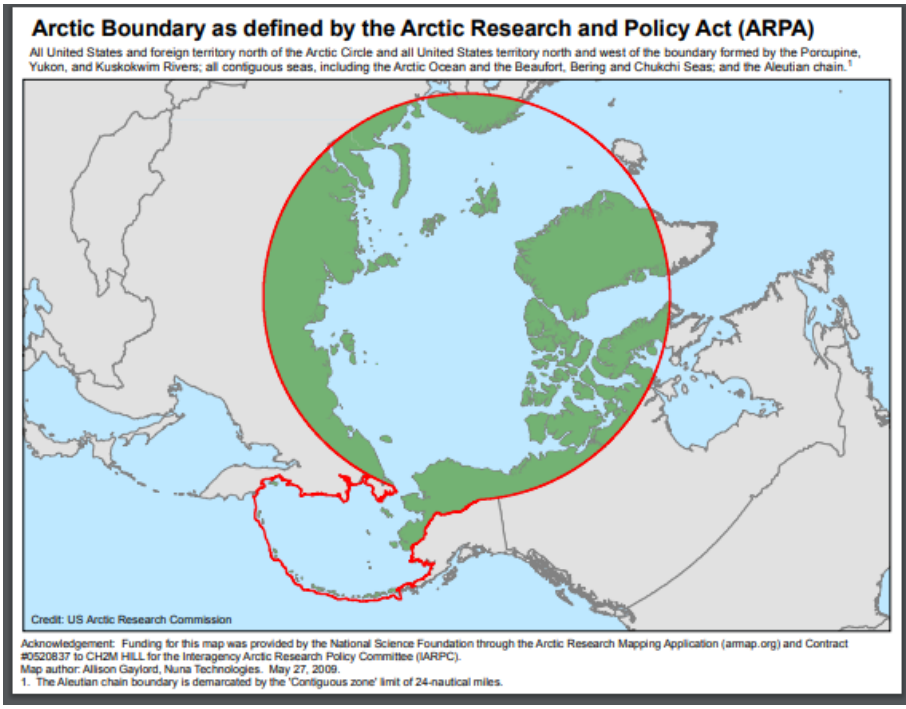


Environmental groups may have motivation to expand the definitional boundaries of the Arctic, in order to encapsulate more of the area under the umbrella of the Arctic Council's working groups. For example, this map depicts the boundaries defined by the Arctic Monitoring and Assessment Programme, one of the working groups of the Arctic Council, whose mandate is to "To monitor and assess the status of the Arctic region with respect to pollution and climate change issues."[4] AMAP acknowledges on their website that their definition of Arctic "extends beyond the Arctic to address global connections associated with long-range transport of contaminants" This holistic view of the Arctic is one that might be popular among environmental circles, but perhaps less practical in terms of cost when considering regulations for those operating in these areas.

The area of the region may also be established via international regulatory agencies. This map shows the IMO definition for polar waters and the area where ships are obligated to abide by the *Polar Code*. Notice here that the Bering region delineation is at 60 degrees North latitude, well above the AMAP boundary, which extends farther south to the Aleutian Island Chain. The IMO definition also excludes Iceland, Faroe Islands and all of Scandinavia (except for Greenland and Svalbard). The IMO's decisions regarding delineation of some regions of the Arctic as compared to others certainly bears further consideration, as it may pose additional problems with consistency of regulatory enforcement for ships sailing in the area.



Alternatively, countries may have motivation to instead define the Arctic in terms that would give the most control over their borders. For example, this map illustrates Arctic as defined by the United States' 1984 Arctic Research and Policy Act. In the statute, the Arctic is defined as "all United States and foreign territory north of the Arctic Circle, and all United States territory north and west of the boundary formed by the Porcupine, Yukon, and Kuskokwim Rivers; all contiguous seas, including the Arctic Ocean and the Beaufort, Bering and Chukchi Seas; and the Aleutian chain."[5] This particular definition of the Arctic includes areas as low as 52 degrees north latitude. [6]



With so many different definitions, it is perhaps unsurprising that some may have difficulty conceptualizing what is meant by "the Arctic." Shifting regions also creates confusion when it comes to overlapping initiatives, regulatory compliance and information sharing.

Our Area of Interest

For the purposes of our report, we focused specifically on the entirety of the Arctic North Atlantic which includes the G-I-UK (Greenland, Iceland, United Kingdom) gap. We also chose to utilize the "above 50 degrees" definition, as it was consistent with the data provided for analysis.



Like all who attempt to study this region, we recognize the variability associated with ecosystems, economies and preparedness for incidents between countries, sub-national regions and waterways. However, by including the entirety of the region, we are able to get a holistic view of cooperation and activity in the region. By widening the scope of the region, themes emerged that may have been missed otherwise if we were to adopt a more narrow geographic focus.

Economic Activity in the Arctic

Different parts of the Arctic and Arctic North Atlantic are home to a variety of different resources for development and exploitation, leading to increased attention to the economic potential in areas of fishing, energy, minerals, and shipping.[7] But the Arctic, like anywhere else, matters differently to different stakeholders, for example, people trying to make a living through economic activity, and groups who have an interest in preserving ecologically sensitive spaces. Because the Arctic spans several different geopolitical regions, it also presents challenges in terms of governance. The following areas represent the largest potential sources for investment in the Circumpolar Arctic.

Fishing

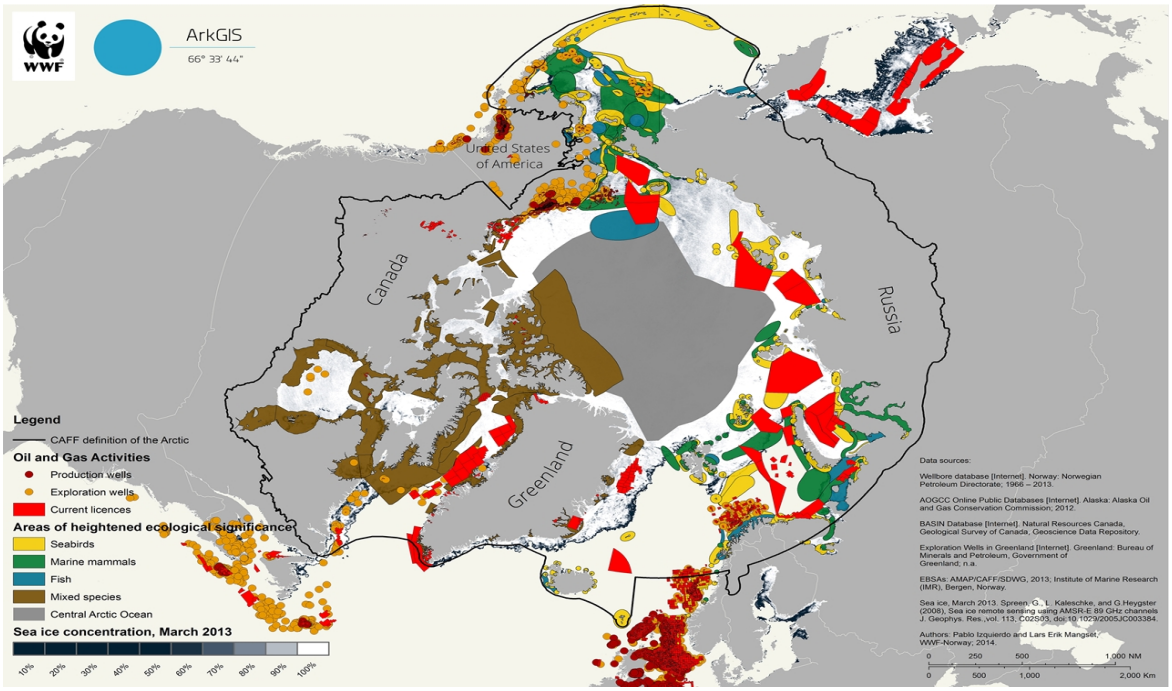
Fishing in the region is responsible for $13 billion dollars-worth of economic activity and represents the biggest source of income for some Arctic states.[8] Signaling how important this resource is to the future sustainability of the Arctic, many of the Arctic states, including the United States and Russia, as well as the European Union and China among others, signed a temporary 16-year moratorium on fishing in parts of the region.[9] The agreement also requires a commitment for participating nations to invest in scientific research of fish stocks as the Arctic climate changes.[10]

Not only is fishing important from a perspective of economic and food security, but fishing in the past has been tied to a country's national security as well. A significant portion of the world's population relies on protein from fish to survive, and differences of opinions regarding a country's right to fish in the past has brought even friendly countries to the brink of war (e.g., the Cod and Turbot wars*).* While there were no combat deaths in these disputes, and they were eventually resolved through the Cold War's necessity for expedient NATO cooperation,[11] it still brings to light that this seemingly otherwise common industry can have implications outside the pier where the fish are unloaded and sold to market.

Mineral Resources

There are also massive reserves of minerals and fossil fuels in the Arctic that may lead to further terrestrial development and increased shipping. A United States Geological Survey report estimates that "the total mean undiscovered conventional oil and gas resources of the Arctic are estimated to be approximately 90 billion barrels of oil, 1,669 trillion cubic feet of natural gas, and 44 billion barrels of natural gas liquids."[12]**.** This would account "for 15 percent of the world's undiscovered oil, [and] 30 percent of the world's undiscovered gas." [13] While some of this discovery is already being pursued by Arctic states and oil companies, the extent to which continued expansion will occur seems at present to be a matter of location. For example, oil exploration in the US Arctic has largely stalled for the foreseeable future after tests showed that the levels of oil and gas were insufficient to warrant further exploration.[14] According to an executive at BP, the Arctic "is a very risky place [to explore] and very expensive to develop, so there are probably easier places to go."[15] On the other hand, Russia's oil and natural gas exploration is booming, and is expected to continue to expand well into the future.[16] Like all activities in the Arctic though, fossil fuel and mineral exploitation can be complicated by the unique and often unpredictable conditions of the Arctic, as well as be influenced by global economic factors, which ultimately makes the prediction of their likelihood difficult.

The map below shows current oil and gas activity, and areas that are licensed for extraction.[17] Drawn by the World Wildlife Foundation, it overlays the oil and gas activity over areas deemed ecologically significant to certain species like fish, seabirds and others.



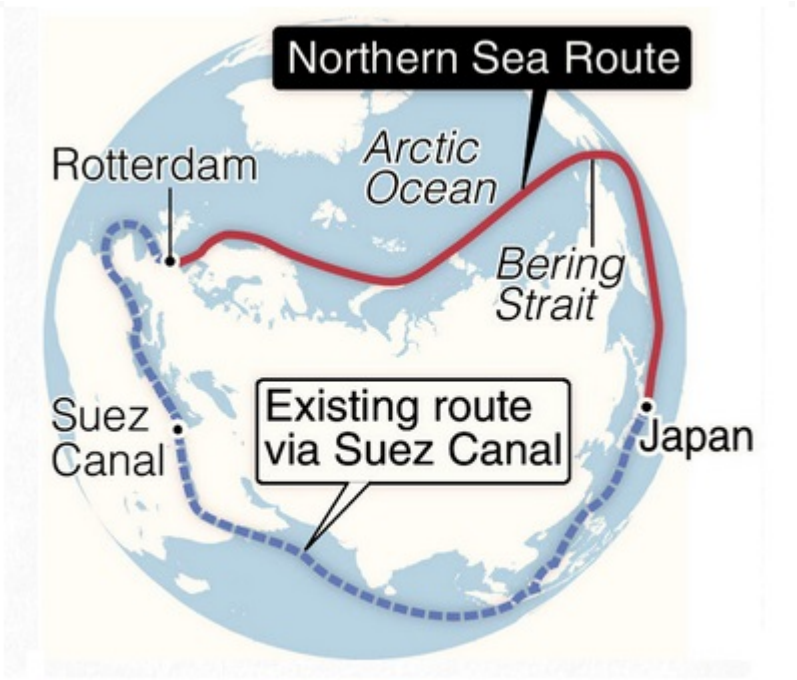
Tourism

Another economic activity thriving in the region is tourism. “Politicians and businesses are hoping and planning for substantial growth in tourism, and with construction of three transatlantic airports planned, a broader societal discussion of how (much) tourism should be developed, in what ways, and by whom, is lacking.” [18] Tourism in the region varies from transatlantic visitors to Arctic countries, to passenger vessels that range from a few hundred to 1500-person cruise ships. As long as humans have had access to these colder, remote regions, they have piqued the curiosity of would-be adventurers.[19] What was normally reserved for the extremely wealthy, or extremely connected has now become available to a growing global middle class. For example, Iceland alone saw a 60% increase in tourists between 2010-2016 according to one database.[20]These increases in the number of tourists is part of why many Arctic stakeholders fear that the danger to the Arctic is increasing, both in the form of over-tourism, and in terms of potential incidents which can occur due to the influx of expedition or adventure tourists.

Commercial Shipping

Commercial container shipping can come in different forms. Distinguishing between these types is important, because there is a belief that one type may increase, while another will stagnant. Destination shipping is defined as that which is “conducted for community re-supply, marine tourism and moving natural resources out of the Arctic.” [21] All indications point to destinational traffic continuing to be the dominant form of shipping in the region, given the number of people moving to and from isolated communities.

Destination shipping can be contrasted with transit shipping. For some, there is a hope that transit shipping would utilize the entirety of circumpolar shipping routes to redirect from warm-water routes that go through the Straits of Malacca, Suez or Panama Canals. A study published by the Asian Journal of Shipping and Logistics found that "if the NSR becomes commercialized, [companies] will be able to save about 5,000 nautical miles in distance and sailing time." [22]



Northern Sea Route and Suez Canal Route

A 2018 headline drew international attention when shipping giant *Maersk* conducted a test of the Northern Sea Route in northern Russia,[23] leading many to speculate that this could be the new norm in global shipping. This speculation has led to a debate among many stakeholders of whether or not this will be the case.

One of the major expectations of shipping is the idea of "reliability." Despite a trend of seasonal ice melt, the environment in the Arctic does not appear to qualify as a viable, or reliable alternative to warmer, more trafficked routes. This is because, according to Palle Laursen, a spokesperson from *Maersk,* "Today, the passage is only feasible for around three months a year,"[24] making it difficult to justify the investment required for ice capable ships for those months. Stephen Carmel, a senior executive at Maersk, also points to the cost-benefit ratio involved in trans Arctic shipping, highlighting the need to "…[formulate] appropriate strategy and intelligent investments, [and the importance of getting] past the hype."[25] According to Carmel, just because there is less sea ice, does not mean that increased transit shipping will follow.

The belief that the Northern Sea Route may become a viable alternative to other global shipping routes is not one that is unwarranted. Those who see this region as a hub of transit shipping in the future point to the substantial recent investments by Russia on their northern border, and to China's desire to become an Arctic player. According to Malte Humpert, a senior fellow at U.S.-based think-tank Arctic Institute, Russia's investment in the region is seen as a point of national pride.[26] Investments by Russia and China to make the region more reliable, coupled with a decrease in sea ice for longer periods of the year,

makes it unclear just how long the regions' transit shipping routes will remain impractical for business.

Arctic Marine Shipping Assessment (AMSA)

In 2009, the Arctic Marine Shipping Assessment was published by the Arctic Council's Protection of the Arctic Marine Environment (PAME) working group focusing on current and future Arctic marine development. In over 200 pages, attention was focused on "natural resource development, governance challenges, climate change and marine infrastructure issues, [and how they were] influencing current and future marine uses of the Arctic." In sum, the focus was on how many ships could be expected to set their sights on the area in the coming years as ice began to recede at the top of the world. Implicit in this publication was the idea that as ice left, shipping of all kinds would rush to take its place, making the Arctic host to the world's new commercial shipping routes. In this work, attempts were made to quantify what was happening in the area, and predict what was to come in an effort to establish new safety standards by which this influx was to happen. Also, a call was put out to preserve and protect the pristine Arctic marine environment, encouraging the responsible development of an area which had up until then remained largely untarnished. At the time it was published, the key findings in the report concluded:

- *"Arctic natural resource development (hydrocarbons, hard minerals and fisheries) and regional trade are the key drivers of future Arctic marine activity.*
- *Arctic voyages through 2020 will be overwhelmingly destinational, not trans-Arctic.*
- *Gaps in hydrographic data exist for significant portions of primary shipping routes important to support safe navigation.*
- *Except in limited areas of the Arctic, there is a lack of emergency response capacity for saving lives and for pollution mitigation, including serious limitations to radio and satellite communications and few systems to monitor and control the movement of ships in ice-covered waters. The current lack of marine infrastructure in all but a limited number of areas, coupled with the vastness and harshness of the environment, makes conduct of emergency response significantly more difficult in the Arctic."*

The report concluded with 17 recommendations meant to address the shortcomings that had been found and attempt to shape what responsible development might look like. To date, two updated reports have been published, in 2011 and 2013 reporting on the progress made towards these recommendations. But progress has been slow, and many of the concerns noted in the AMSA remain, even some 10 years later.

Global Commons versus National Sovereignty

With the awareness that there is economic value to be derived from the Arctic/North Atlantic comes a discussion of to whom the resources belong. One attempt to create an orderly process to claim these resources was through the *United Nations Convention of the Law of the Sea (*UNCLOS). UNCLOS is an "international agreement that defines the rights and responsibilities of nations with respect to their use of the world's ocean." Since the ratification of UNCLOS by 166 countries, four of the Arctic states have made extended territorial claims in the region: Canada, Russia, Norway and Denmark.[27]

Making territorial claims now may establish authority and sovereignty over each region's economic resources as states and industry move into the area in the future. Worth noting, is that of all the Arctic nations, only the United States has not ratified the UNCLOS agreement, making it unclear how potential disputes over the US Arctic will be handled or resolved in the future.

Arctic Governance and Cooperation

> *"Eight countries make up the Arctic: The Kingdom of Denmark, Iceland, Norway, Sweden, Finland, Russia, the United States and Canada. And nothing would be worse than each of these countries trying to exploit the maximum instead of working with the other countries to promote responsible stewardship. Nothing could be worse than militarization based on mistrust between these countries that are neighbours." -- Pamela Goldsmith-Jones, Parliamentary Secretary to the Canadian Minister of Global Affairs, 29 September 2016*[28]

At present, governance of the Arctic happens through a series of intergovernmental fora, each having responsibility for oversight of different aspects of operation.

The Arctic Council

In an effort to prevent chaos and destructive self-interest in the Arctic, the Arctic Council was created to be a "high level intergovernmental forum [which] provides a means for promoting cooperation, coordination and interaction among the Arctic States."[29] The Council makes distinctions between three types of groups; member states, permanent participants and observing states, each with different qualifications and obligations.

The council is comprised of the eight Arctic states, the United States, Russia, Norway, Sweden, Finland, Iceland, Canada and the Kingdom of Denmark (Greenland). Membership was outlined in 1996, in the *Ottawa Declaration,* the founding document of the Arctic Council. All these countries have some territory inside the Arctic Circle, or just above 66 degrees north latitude, which makes their stake in the international governance of the region clear. The other members of the Arctic Council include permanent participants which are made up of six Indigenous groups who live in the Arctic and have a stake in how it is governed economically and politically.[30]

The Council also allows for the presence of other governmental entities in the form of observing members. Observing members are those which have no territorial claim of lands in the Arctic, but may have an interest, morally or economically, in the development of the Arctic. Observing members are sub-Arctic countries like China, Korea and the UK. They may also be inter-state actors like the Nordic Council of Ministers and The United Nations Environmental Programme or non-government agencies (e.g., the Circumpolar Conservation Union and National Geographic Society). These observing members can make statements, issue recommendations or share expertise in the region.[31]

Much of the work of the Council happens through the efforts of six working groups, including: Arctic Contaminants Action Program (ACAP), Arctic Monitoring and Assessment Programme (AMAP), Conservation of Arctic Flora and Fauna (CAFF), Emergency Prevention, Preparedness and Response (EPPR), Protection of the Arctic Marine Environment (PAME), and the Sustainable Development Working Group (SDWG).[32] All of these working groups are funded directly by Arctic Council nations, and work usually on a project-to-project basis for research and event hosting.[33] The scope of our project most closely aligned with the work of the EPPR, although in many cases, the missions and work of the other working groups often intersect. This is because the EPPR, while having no operational capacities themselves, organize and conduct research for environmental protection and best practices for those who operate in the region.[34]

Most notably, the EPPR has facilitated two landmark agreements between the states on the Arctic Council, the Search and Rescue Agreement and the Agreement on Cooperation on Marine Oil Pollution Preparedness and Response in the Arctic (MOSPA).[35] Since the agreements were made, Arctic Council countries meet biannually to plan exercises, discuss best practices and establish contingency plans.[36] (In fact, part of our research involved viewing these exercises to identify some of the lessons learned to frame policy recommendations.)

By agreement, the Arctic Council is limited to political and regulatory discussions only, and cooperation is limited to non-military affairs, which hints at the notion of diplomatic isolation for issues facing the Arctic.[37] But issues regarding coastal protection and security affairs, instead are under the purview of the Arctic Coast Guard Forum.

The Arctic Coast Guard Forum

The Arctic Coast Guard Forum (ACGF) is another venue by which international cooperation is facilitated in the region. Separate from the Arctic Council, and not being born out of a treaty, its chairmanship and members are the same as the Arctic Council with the exception of the Indigenous permanent participants and observers. The group meets semi-annually with a focus, as the name implies, on Coast Guard-specific operations, focusing on best practices to and how to best achieve cooperation in the region.[38] Like the Arctic Council working groups, the ACGF hosts live and table-top exercises regularly that can be examined to identify lessons learned.

Ilulissat Declaration

In addition to these other cooperative fora, the *Ilulissat Declaration* also gives reason to believe that there will be continued cooperation in the region. This agreement, signed in 2008 by United States, the Russian Federation, Canada, Norway and Denmark, enumerated the countries' intentions' to resolve conflicts in the region through diplomacy. It also prioritizes scientific research, sustainable development and safety of navigation.[39] The fact that this agreement has remained in place for 11 years serves as an example to those who believe that the level of cooperation in the Arctic is exceptional, despite the United States' failure to ratify UNCLOS.

Risks in the Arctic

As implied by the two preeminent agreements to come out of the Arctic Council, areas regarding search and rescue, and environmental protection represent two of the most

pressing risks to the pristine Arctic ecosystem. This region involves more risk than most others in the world. Even though the sea ice is receding, ice floes, dangerously cold weather, and uncharted waters characterize the dangers of this newly opening frontier. Even though industrial development represents much in the way of promise for the region, it also brings impending environmental pollution, overtourism and threats to Indigenous peoples' way of life.

Search and Rescue

One of the biggest concerns with regard to industrial development in the Arctic relates to the vast distances embodied in the Arctic, when compared to the existing infrastructure and capabilities to respond to accidents and incidents. By many, if not all accounts, the infrastructure that is there is sufficient only to handle the small numbers of locals who live in the region, and would quickly become overwhelmed with even small incidents which may occur in the region.

Along with limited infrastructure in terms of hospitals and medical response capabilities on land, is a lack of permanent rescue assets like icebreakers, long range helicopters and in some cases, sufficient crew. In 2007, a small expedition style cruise ship called the *Explorer* sank in the Arctic after striking an iceberg, requiring the rescue of 150 passengers and crew. Even though none of the passengers or crew perished in the incident, it took over 5 hours for them to be rescued from lifeboats. By standards of response in the region, 5 hours can be an incredibly short amount of time, because depending on where a potential incident occurs, help may be days away due to the lack permanent assets in the region.

Oil Spill

Many of the same assets that would be deployed for search and rescue operations would be required for oil spills. While oil spills in any marine environment can be costly and difficult, like search and rescue operations, the challenges inherent in the Arctic region like cold weather, poor satellite communication, poor visibility, and the lack of infrastructure make the threat of an oil spill even more problematic.

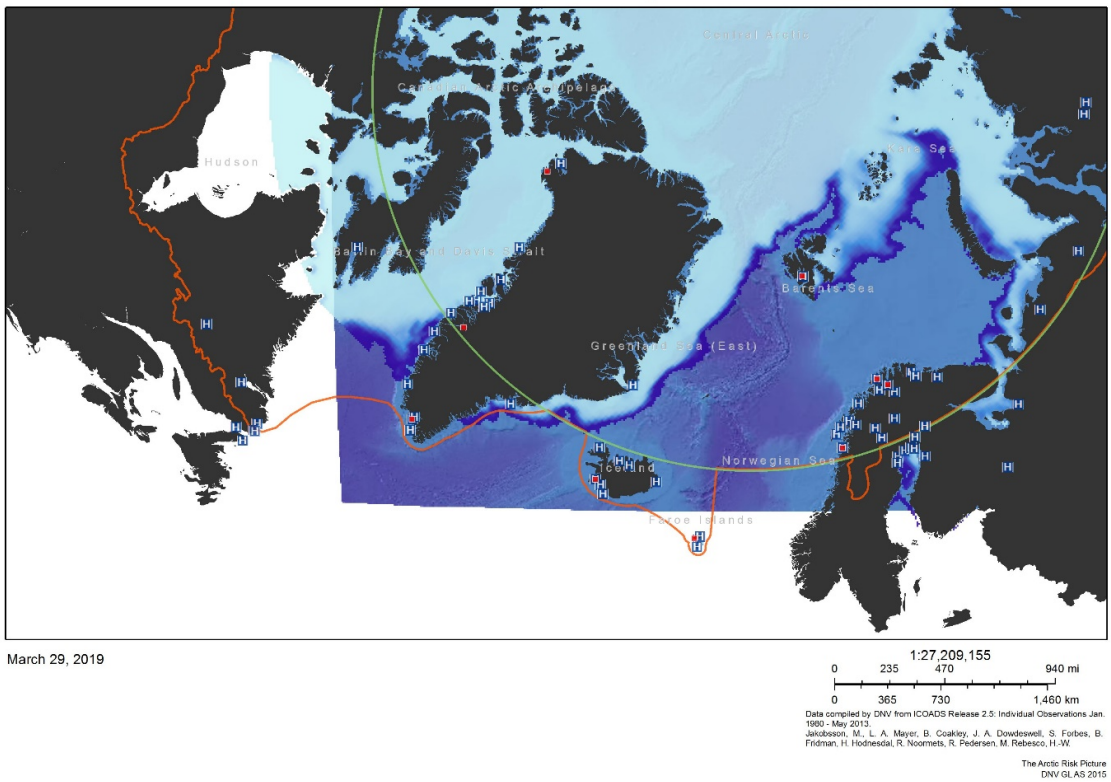
The unique nature of the region, because of the presence of sea ice, can also lead to challenges for oil spill response as compared to other areas where oil is extracted and transported. According to the Arctic Oil Spill Response Technology Joint Industry Programme (JIP), an organization of nine oil and gas companies that examine risks and best practices associated with fossil fuel extraction and usage in the region, oil behaves differently in Arctic environments.[40] Because of oil's unique behavior, this also influences how it can be cleaned, and at present, there is insufficient knowledge as to how to clean up oil in ice.[41] For this reason, continued research into the unique risks posed by oil and gas exploration and extraction in Arctic regions should continue, so that both industry and first responders are able to assess risk and develop strategies for deploying assets in the region.

Communication and Cooperation

To help give a sense of what cooperation actually looks like on a practical, operational level, the map below, created by the risk-management company DNV GL, shows the current location of helicopter bases (red) and hospitals (blue 'H') that would be activated

in the event of an emergency in the region. The vast distances from helicopter bases to remote areas is notable, as well as the expanse of areas with no hospital facilities. Looking at this map, it is easy to see why so many are concerned about sufficient SAR resources.[42]

The Arctic Risk Map

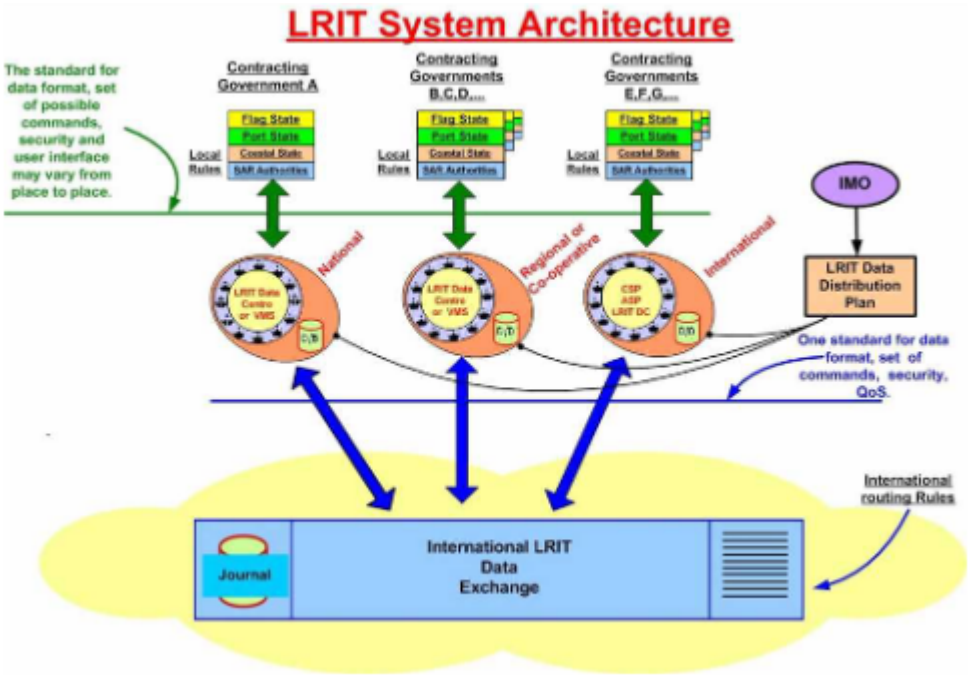


An additional issue that arises when considering communication and international cooperation in the region stems from the fact that search and rescue resources are, for some Arctic states, intertwined with military capabilities. Enmeshing search and rescue with national security concerns means that information sharing may not be as simple and straightforward as it otherwise could be. The United States, Norway, and in some cases Russia (whose coastal responsibilities are shared between military and civilian offices) have Coast Guards that operate under military organizations.[43] Canada and Sweden, on the other hand, have Coast Guards that function under civilian departments.[44,45]Whereas no one could reasonably make the claim that any deployment of military ships for the purposes of search and rescue warrants fear of *militarization*, it could make coordination in the form of information sharing much more difficult. The fact also remains that there have been concerns raised that Russia's buildup of infrastructure in the region could have hard security implications, in ways that buildup of civilian Coast Guard resources might not.[46]

In addition to coordination among Arctic states, are issues with communication and coordination among Arctic states and other non-governmental entities. It was only relatively recently (i.e., 2000) that the IMO mandated that ships utilize ship tracking systems like AIS, to facilitate awareness of ship locations by coastal authorities. Other

initiatives for vessel tracking followed, including the creation of databases for vessels of opportunity for emergency response cooperation. Some examples are the voluntary Atlantic Merchant Vessel Emergency Reporting (AMVER) System based out of the United States, and the IMO mandated GREENPOS out of Greenland. The goals of these databases were to effectively “"take the search out of search and rescue” by ensuring that routes were planned and communicated to rescue agencies in the event of an incident.[47] These are examples of responsibility being placed on industry to be proactive in the case of emergencies where every moment lost could be catastrophic.

Another recent example of attempting to organize ship traffic information is in the Long Range Identification and Tracking (LRIT) system. Below is an image that shows how the LRIT was created by the IMO in an attempt to ease information sharing for governments and emergency responders.[48] Particularly worth noting is the requirement of a standardized format to ensure vital information is accessible across borders.



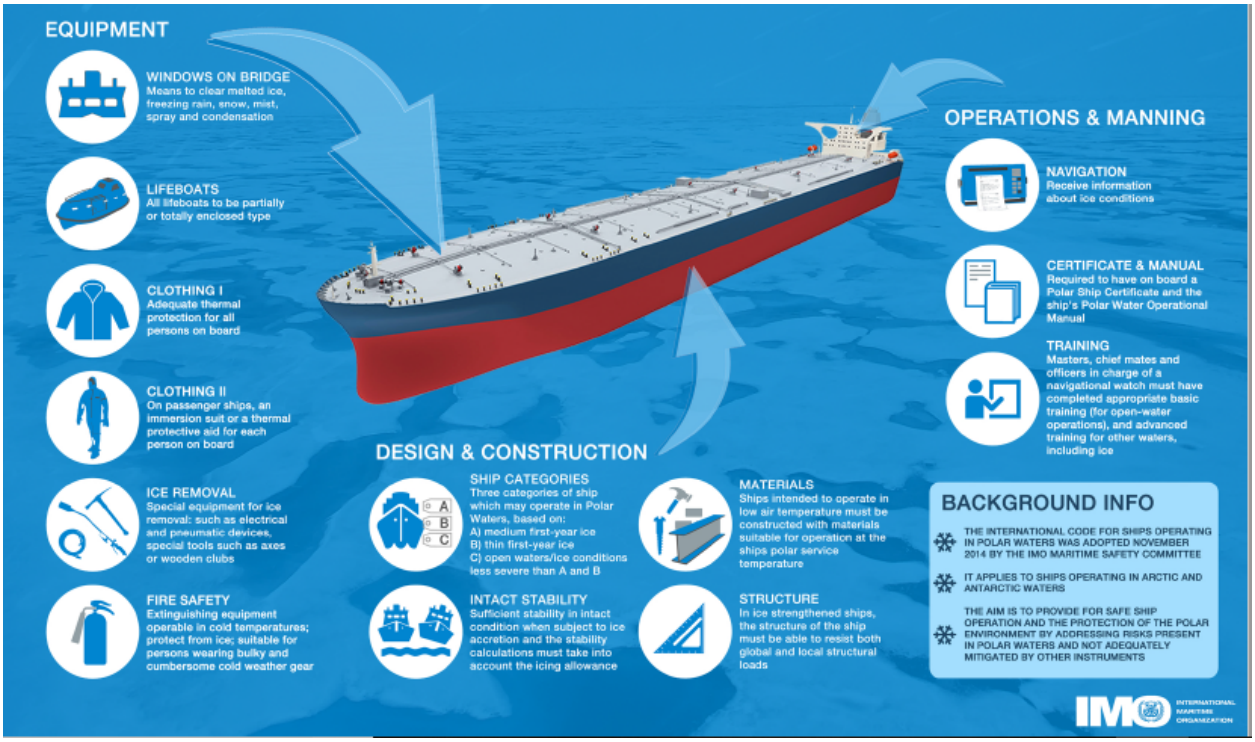
The importance of these databases to all stakeholders in maritime activities is clear, but especially in the Arctic where emergency responders could take longer to meet those in distress due to the vast distances and remote locations.

The Polar Code

In response to some of these unique challenges and increased risks inherent in operating in Arctic and Antarctic waters, the IMO also put into place a series of other mandated requirements. The International Code for Ships Operating in Polar Waters (Polar Code) represents a series of regulations that “protect ships and people aboard them, both seafarers and passengers, in the harsh environment of the waters surrounding the two poles.” [49]

The Polar Code presents a combination of functional safety regulations and environmental protections. Some of them are mandatory, like requiring cold-weather gear for crew and passengers,[50] and banning of dumping certain materials deemed environmentally hazardous[51]. The Polar Code also requires ship owners to specify their ship's unique limitations and capabilities, as well as form a plan for what to do if an incident were to occur in polar waters. The Polar Waters Operational Manual is required by the Polar Code, and it requires ships to specify their voyage plan, as well as "risk-based procedures to be followed for contacting emergency response providers for salvage, search and rescue (SAR), spill response, etc." [52] They also must include "procedures for maintaining life support and ship integrity in the event of prolonged entrapment by ice, and shall include risk-based procedures for monitoring and maintaining safety during operations in ice, as applicable, including any requirements for escort operations or icebreaker assistance."[53]

The graphic presented below highlights some of the mandatory requirements put in place by the Polar Code, required for all new ships built after 2017. [54]

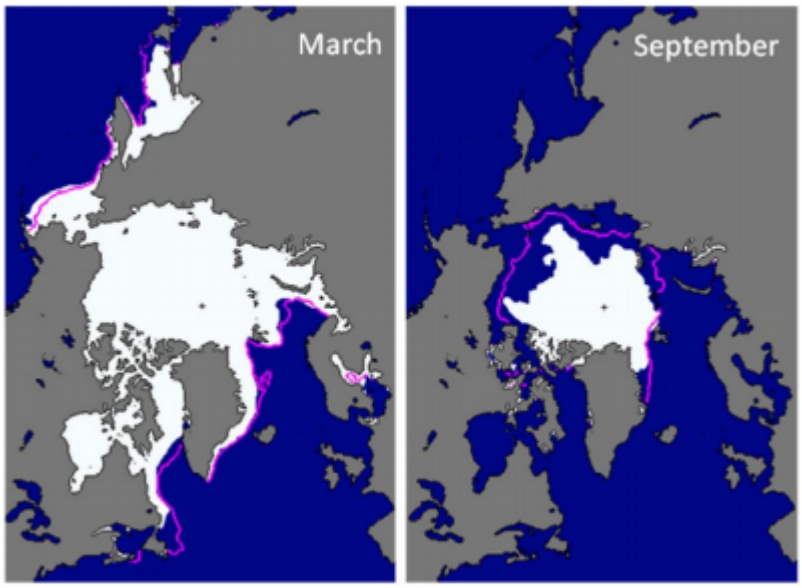


Limits of The Polar Code

The boundaries to which the Polar Code applies, along with the provisions, are the result of years of discussions and debates among stakeholders. Those debates, according to some Arctic stakeholders, left a lot to be desired or overstepped the bounds necessary to govern the region safely. Worth highlighting are some of the geographic limits of the Code, and questions regarding enforcement.

For example, some have questioned the decision to include the entire island of Greenland, while excluding all of Iceland, and most of coastal Norway, given that

Norway accounts for approximately 80 percent of all maritime traffic in the Arctic. One justification for these decisions was the belief that "Sea ice extent should be the primary determinant of the Polar Code boundary in the Arctic."[55]



From data made available by the National Oceanic and Atmospheric Administration, the boundaries do appear to be consistent with such a definition for the region.[56]

However, the presence of permanent sea ice might not be the best determining factor of what should be included in laws governing polar waters, especially since levels of sea ice are changing, and many of the same weather conditions and dangerous ice floes can be found in places outside the geographical delineations. Not requiring vessels sailing in similar conditions to adopt the safety regulations in the Polar Code may leave them at risk for the very problems the code was trying to address.

In addition to geographical limits of the Polar Code's application, were limitations in terms of the vessels to which the Code applied. The IMO's Safety of Life at Sea (SOLAS) agreement, of which the Polar Code is an amendment, does not apply to vessels under 150-500 gross tonnage (GT), and instead delegates authority to flag states on how to regulate these vessels depending on the nature of their route.[57] The decision to exclude these vessels from the regulations of the Polar Code makes these smaller vessels a higher risk, especially considering the nature of these types of vessels, fishing vessels and smaller pleasure craft. It is difficult to say why the Polar Code does not include these types of vessels except to say that fishing vessels are not covered under SOLAS either. According to one IMO webpage, "The safety of fishing vessels has been a matter of concern to IMO since the Organization's inception, but the differences in design and operation between fishing vessels and other types of ship have proved to be an obstacle to their inclusion in the SOLAS…Conventions."[58] It would appear then that regulations with regard to these types of vessels is lacking as compared to other types of ships sailing in the area.

The issue of enforcement is one that remains as a criticism of the Polar Code as it exists now. Some have pointed to the difficulties that flag states have in ensuring compliance with the Code. This is a concern mirrored by classification societies as well who have struggled with how to ensure compliance with the code in light of its goal-based nature.[59]

It is also possible that some ships may be motivated to seek out particular flag states in an effort to avoid strict Polar Code enforcement. In a report examining issues of flag-state compliance, Matthew Gianni, in conjunction with the World Wide Fund and the International Transport Workers' Federation, found that, "to shipping and fishing interests, the economic attraction of registering a vessel under a flag with lower costs, more relaxed crewing requirements and less vigorous regulation is undeniable.[60] The Polar Code does not address this issue of flag state compliance, which can have extremely high consequences when disaster strikes.

The Current Study

In conclusion, it is clear that the Arctic is a rich and varied area where many different stakeholders' interests and capabilities combine. Even the word Arctic does not mean the same thing to all people. No matter how you define it, however, the region is not isolated from the rest of the world, where geopolitical and economic power influence behavior.

As mentioned earlier, the goal of the current study was to prepare a foresight analysis regarding issues and incidents likely to occur in the Arctic North Atlantic region in the coming years. We also hoped to examine any policy gaps that exist in regulations of Arctic Shipping. Since development in the Arctic is still relatively new, we felt the best approach to understanding the future of Arctic shipping was to include factual data from historical events, insight from Arctic expert stakeholders, and an examination of available documentation in the form of after-action reports from joint exercises.

We adopted a rather liberal definition of the Arctic for the purposes of our study, choosing to examine incidents north of 50 degrees latitude, and encompassing the entire area between Canada and Norway. Although these decisions may not adhere to others' more traditional definitions of what constitutes the Arctic, we felt it would present the most inclusive picture of what we can expect in the region over the next 15-20 years.

Historical Trends in the Arctic as a Whole

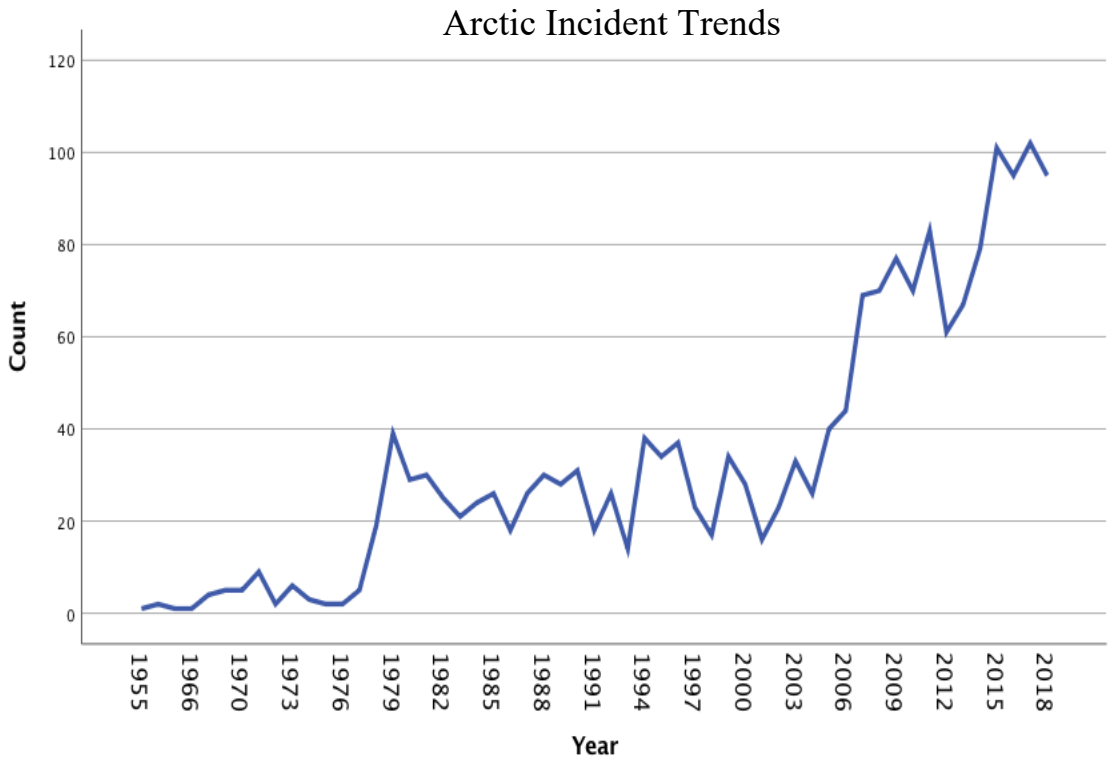
In order to forecast the likely casual factors of shipping incidents in the Arctic over the next 15-20 years, we examined historical trends in marine accidents and incidents occurring in the last 64 years. Casualty data was provided from IHS Lloyd's Register Fairplay.[61]. The data included 1822 casualty incidents (from 1552 ships), occurring above 50 degrees latitude from 1955-2019, but also provided details regarding ship characteristics (year of build, ship size, ship type), incident characteristics (year, region, description) and owner/operator information. For the purposes of the current project, only the ship and incident characteristics were examined. There were some mistakes in the database noted (e.g., ship type characterizations varied for the same ship involved in more than one incident, or coordinates which did not match with incident descriptions), so the authors are unable to guarantee the completeness or accuracy of the data upon which the following analyses were based. However, assuming that, on the whole the data represented a representative snapshot of the types of incidents that have occurred in the Arctic region over the last 64 years, quite a few trends emerged that are worth consideration.

Additionally, there were a few issues with the data that was included which made some of the analyses and interpretations more challenging. There was wide variability in the types of information available about each incident, the biggest example of this being with regard to the location of the incident, which was especially problematic when it came to analyses concerning specific regions (e.g., the Arctic North Atlantic). Therefore, these regional analyses should be interpreted with caution as they were conducted only on the subset of the data that specifically identified region. There may have been a higher number of incidents in the area that we were unable to include due to the lack of information provided. The amount of information available as far as descriptions of incidents also varied greatly, which placed some limitations on the ability to compare across incidents, both for the entire Arctic, and the ANA Region specifically.

An additional caveat needs to be included as well. We were unable to obtain data on actual shipping traffic which would allow us to contextualize our findings in terms of the number of ships in the region. This information is important when considering the types of trends that emerged. More specifically, without this information, it is unclear whether increased incidents involving certain types of vessels were due to an increased threat this type of vessel may pose, or was merely a function of increased representation in the population of vessel traffic. For this reason, we attempted to contextualize our findings where possible, but future research examining these trends in light of vessel traffic is important to provide a more complete picture of incidents in the Arctic.

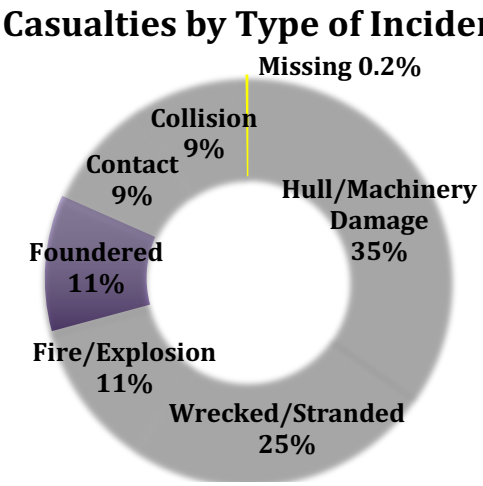
Arctic Casualty Trends

Over the years, there has been a definite increase in the number of incidents occurring in the Arctic, which likely corresponds to the increase in accessibility to the area.



Arctic Casualty Types

All causalities in the Fairplay data were categorized as one of six types: Hull or Machinery Damage, Wrecked/Stranded, Fire/Explosion, Foundered (Sank), Contact, Collision and Missing. Descriptive analyses indicate that the most common incident in the Arctic involved damage to Hull or Machinery, and the least common was for a ship to go missing.

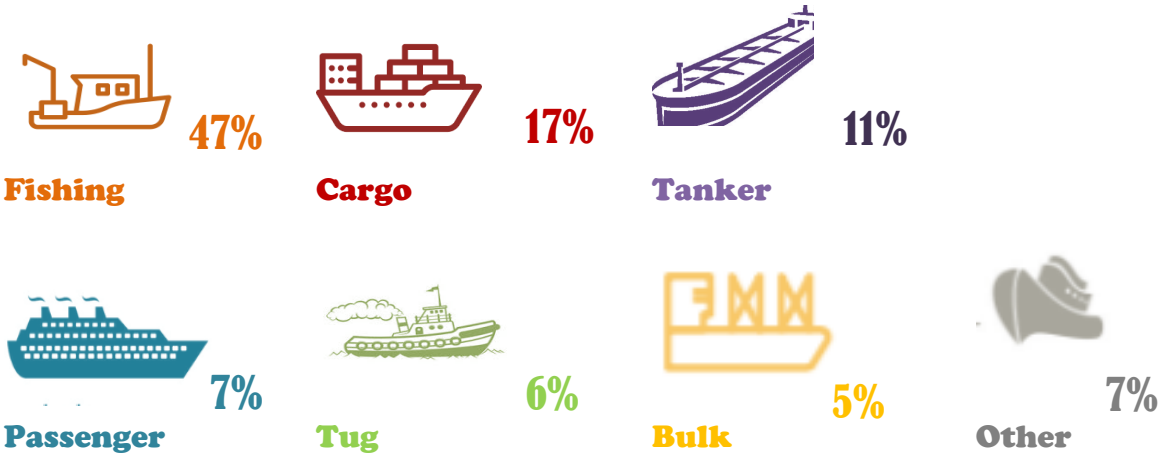


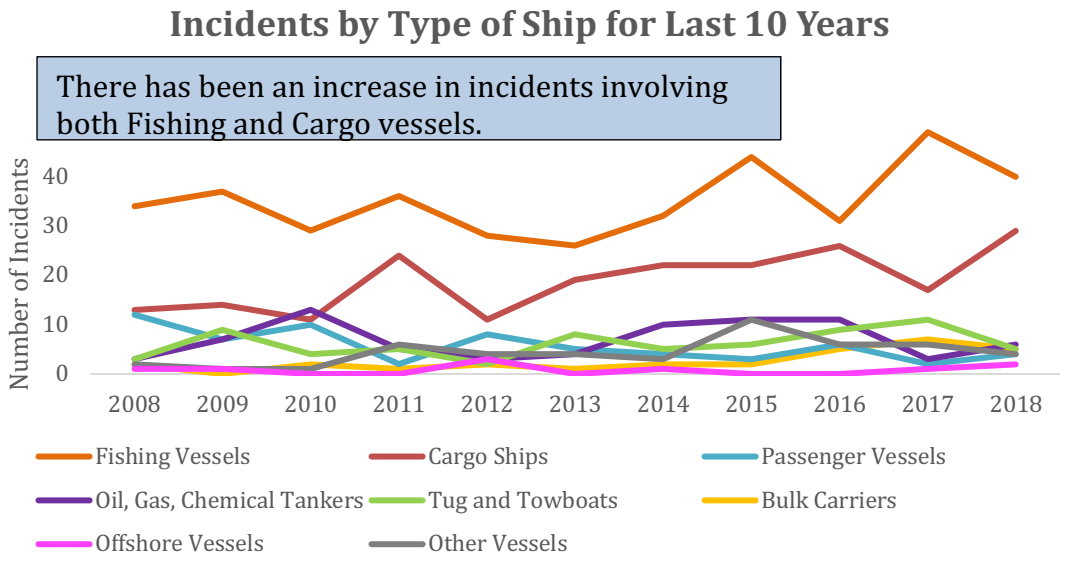
The most common type of casualty was damage to hull or machinery, followed by wrecked or stranded.

According to The Allianz Safety and Shipping Review 2018, worldwide from 2008-2017, the largest cause of accidents/incidents was foundering (n=593), followed by Wrecked/Stranded (n=229).[62] Therefore, perhaps unsurprisingly, there is a greater likelihood of hull and machinery damage in the Arctic region as opposed to the rest of the world.

Type of Ships Involved in Arctic Incidents (1955-2019)

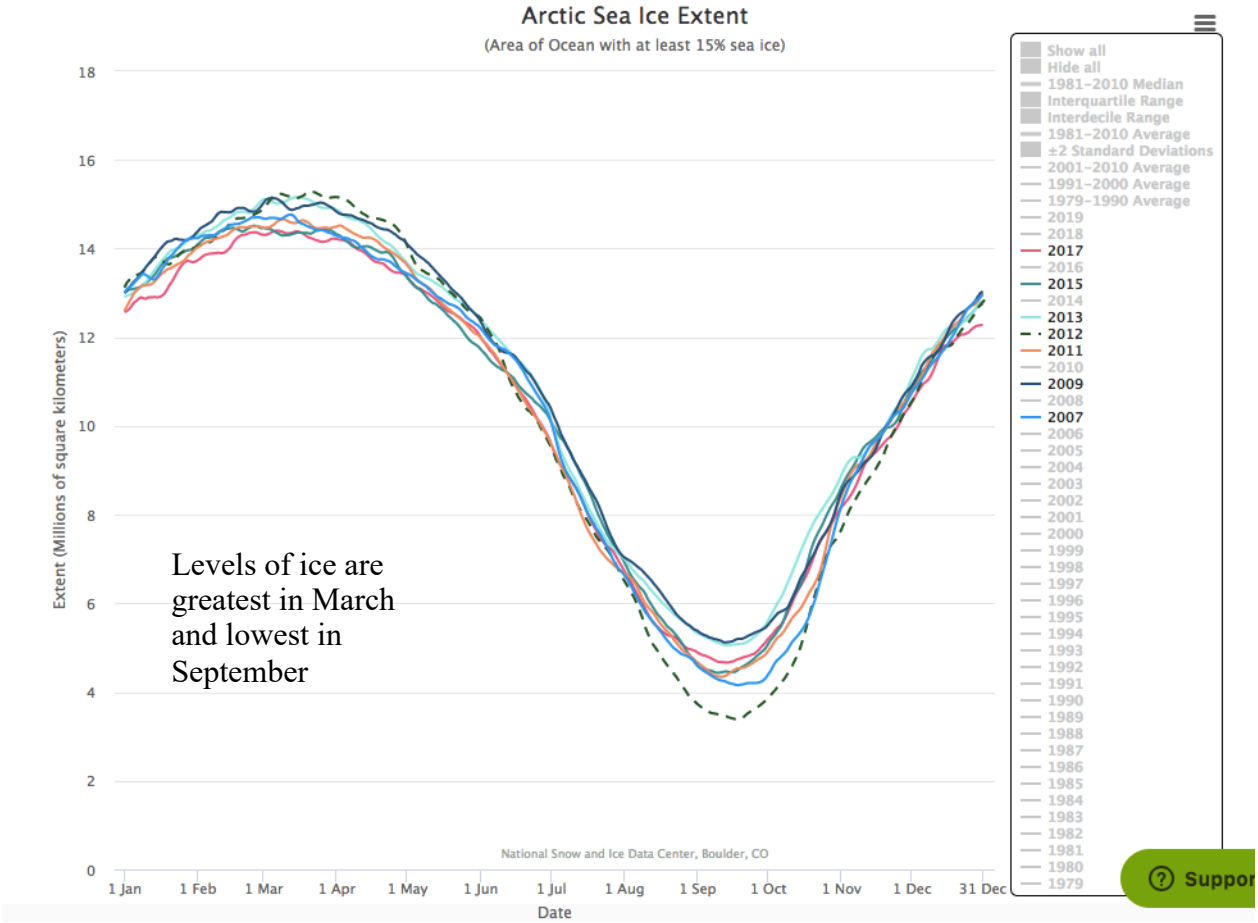
By far and away the greatest number of incidents in the full dataset occurred to fishing vessels, followed by cargo ships and oil/gas/chemical tankers. In 2009, it was predicted that fishing vessels represented 27% of the vessels sailing in the Arctic[63]. According to the incident database, these vessels account for almost half of all shipping incidents.





The variability in incidents over the last ten years poses interesting questions as to the reasons behind these fluctuations. One plausible hypothesis may be that these variations are tied to levels of sea ice. However, data from the National Snow and Ice Data Center

(NSIDC) [64] show that the relationship between incidents and sea ice levels are not as clear cut as they may seem. For example, as shown in the figure below, levels of sea ice were the lowest in 2012 and the highest in 2009 and 2013. While there was an uptick in fishing vessel incidents in 2009, other types of ships did not show similar increases during that time. Further, incident rates for fishing vessels were the lowest in 2013, which was a year that showed similar ice coverage.

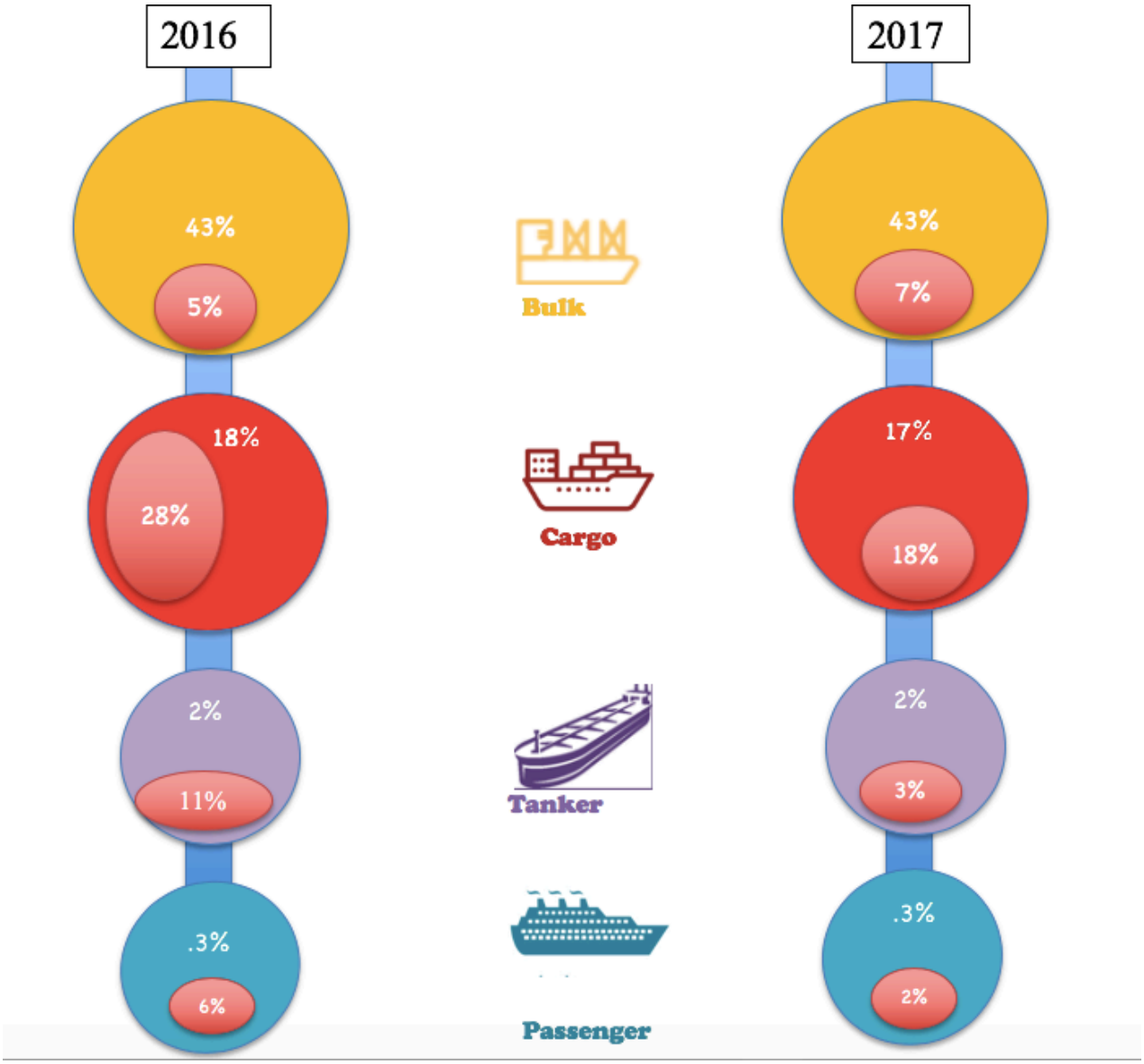


Comparisons to the Worldwide Fleet

According to the Allianz Safety and Shipping reviews, worldwide from 2008-2017, the most common type of vessels involved in casualties were cargo vessels (n=471), followed by fishing vessels (n=174), which is the opposite trend as found in the Arctic. Therefore, it would appear that fishing in the Arctic is much more dangerous compared to the rest of the world.

In order to contextualize the incident rates, we compared the percentage of total ships of each type in the worldwide fleet[65], to the percent of incidents in the Arctic. In the two years for which data was available, Cargo Ships and Oil, Gas and Chemical

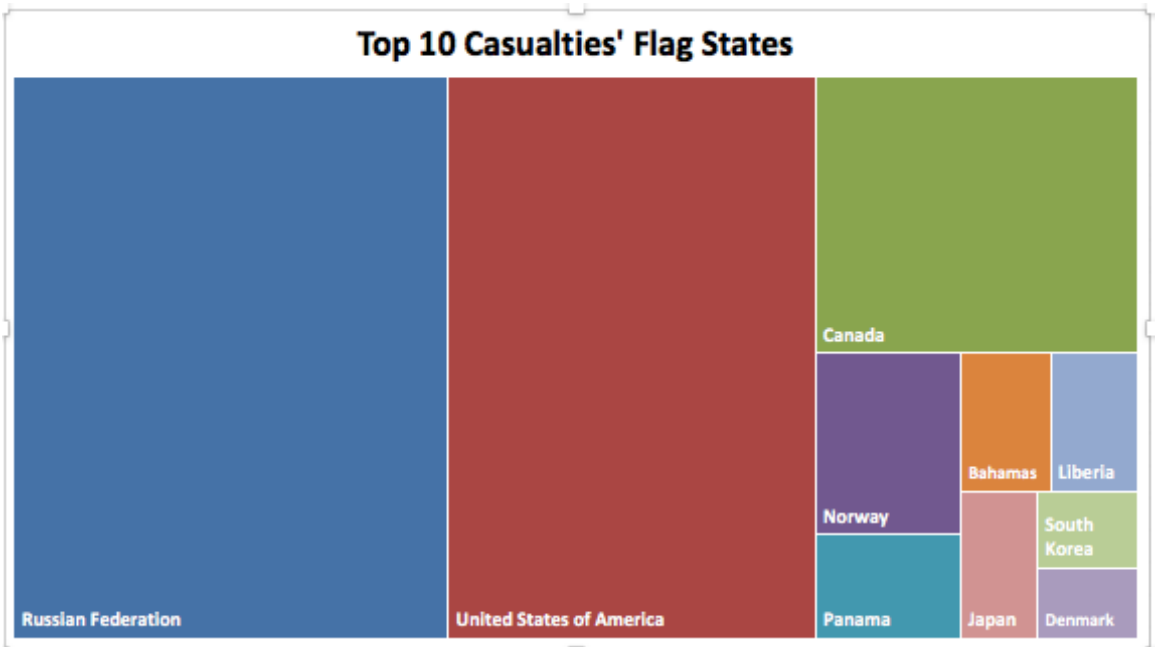
Tankers had a greater percentage of incidents in the Arctic. For example, in 2016, bulk carriers represented 43% of the worldwide fleet, but only 5% were involved in an incident that same year.



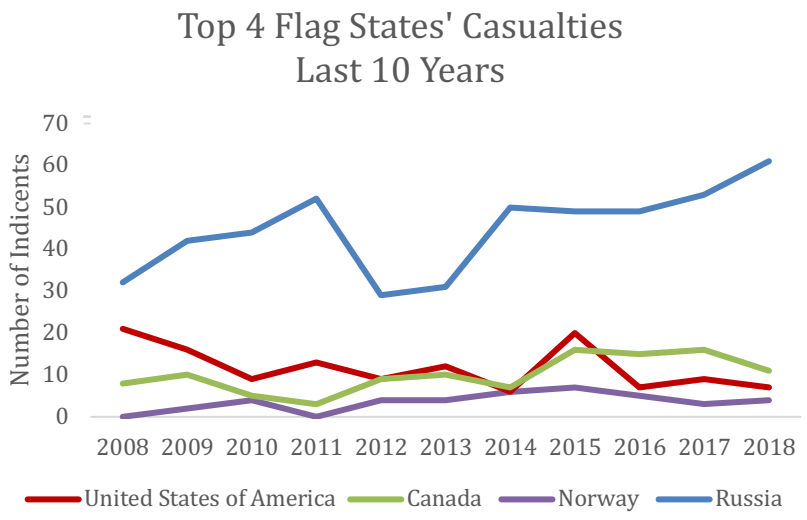
(Data unavailable for other types of ships)

Arctic Incident Flag States (1955-2019)

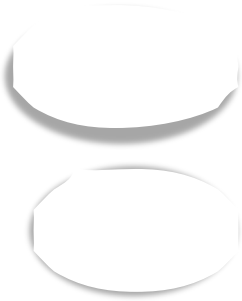
There were 51 flag states represented in the Fairplay data. The top four flag states having the highest number of casualties were coastal Arctic states.



Examining the trends over the last decade, incidents involving Russian flagged vessels have increased, with the number of incidents in 2018 being nearly double what it was in 2008.



Ownership of the World Fleet: Number of Ships 2017[3]

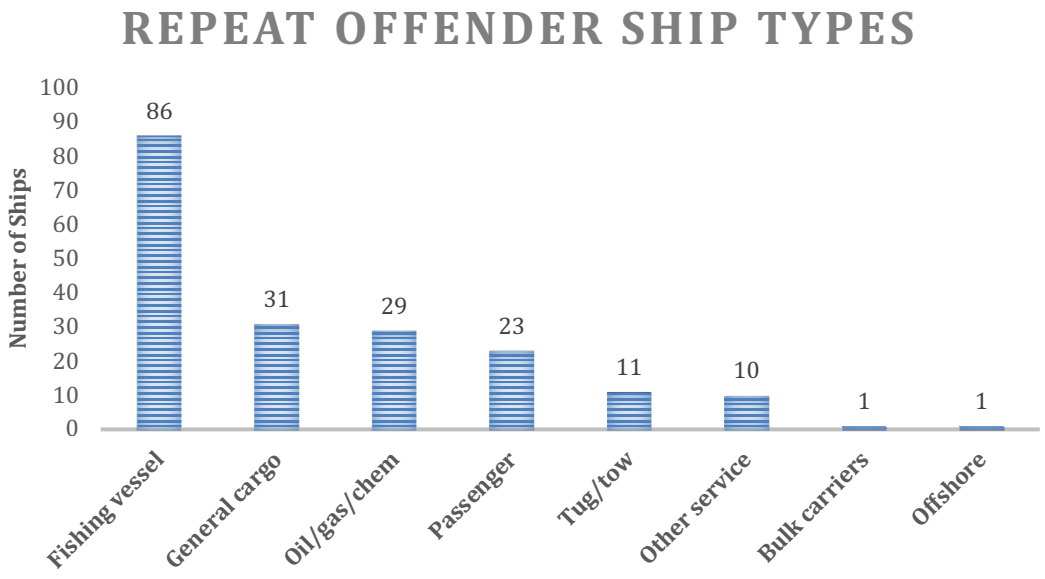


Repeat Offenders

Most ships in the incident database were involved in only one incident. However, 12.5% of ships were involved in more than one incident, with three ships involved in 9 or more incidents.

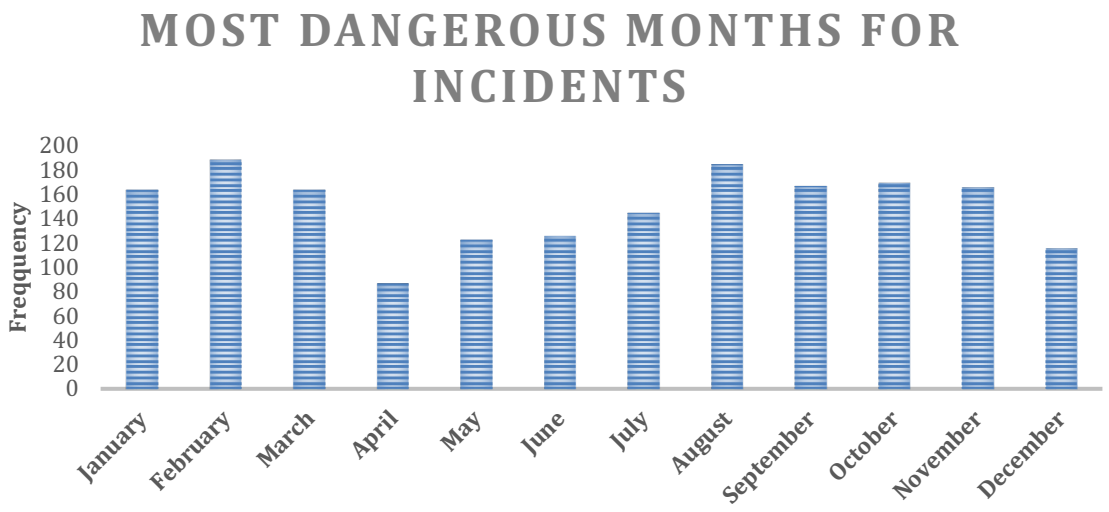
Number of Incidents per Ship	
	1358
	156
	22
	10
+	6
Total	1552

Similar to the overall rates of incidents by ship type, the most common type of ship in the repeat offenders database was fishing vessels, followed by general cargo and oil/gas/chemical tanker.

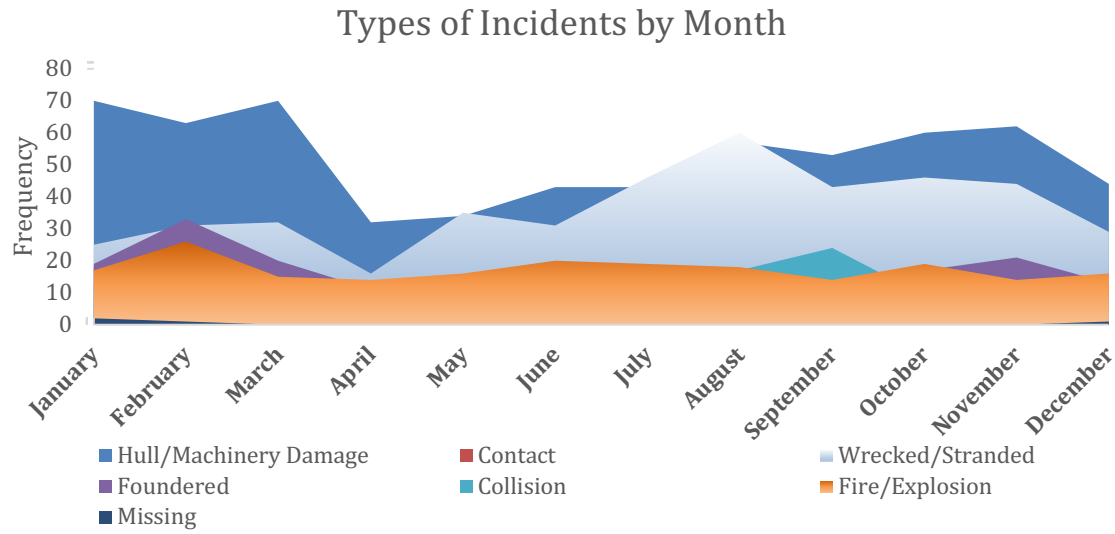


A Timeline of Incidents

The highest number of incidents occurred in February and August, followed by a roughly equal number of incidents occurred in the months of January, March, September, October and November (average =167). The fewest incidents occurred in April.



The bimodal nature of the distribution of incidents likely underscores the role of additional ice in February, and the possible role of increased ship traffic in August. According to the NSIDC, levels of ice are highest in March and lowest in September. This is borne out by the data in that, in addition to the higher risk of hull/machinery damage overall in the arctic, there are a greater number of incidents of fire/explosion and foundering in February, and a greater number of incidents due to wrecked/stranded and collision in August.



Search and Rescue Operations in the Arctic

Only 302 of the 1822 incidents in the casualty database described the need for rescue operations. For those descriptions including the information, number of passenger/crew and type of rescue was examined, along with the vessels and vehicles responding. In general, it seems that the Arctic is relatively safe given that in the last 64 years, only 282 individuals have died as result of marine casualties in the region, with an additional 486 individuals listed as missing. Even assuming all those listed as missing perished, the total number of deaths in the region would be 768 over the last 64 years. (To put this number in perspective, in 2015, the International Transport Worker Federation reported that an average of 2000[66], individuals lose their lives each year in shipping incidents worldwide.) An additional 208 individuals required medical attention at the time of rescue (not including those who died). On the other hand, 3446 individuals were listed as rescued safely (the actual number of which is higher due to the number of entries simply stating "all crew" were rescued safely without providing numbers).

Text entries for casualty events revealed that other, non-Coast Guard vessels were the most common response vehicle (n=96; e.g., fishing vessels, merchant vessels, tug and other vessels), with fishing vessels accounting for the majority of these vessels of opportunity (n=47). Helicopter rescues also accounted for a large proportion of the rescues in the region (n=55). Finally, life rafts and lifeboats were involved in 48 rescue incidents. As is evident from the Figure below, only a very small number of incidents involved icebreakers, or water rescues.



**NOTES: incidents often involved more than one type of asset; and not all entries listed the types of assets involved in rescue operations or utilized the same terminology when listing the rescue vehicles.*

Cruise Ships in the Arctic (1955-2019)

On the whole, there have been very few cruise ship accidents in the arctic (n=54) resulting in 5 deaths and 13 individuals missing (out of at least 8535 passengers where details permitted a passenger count). From these incidents, an additional 97 passengers required rescue medical services resulting from a cruise ship incident. (There have been an additional 5 deaths and 15 missing from incidents involving passenger/Ro-Ro vessels

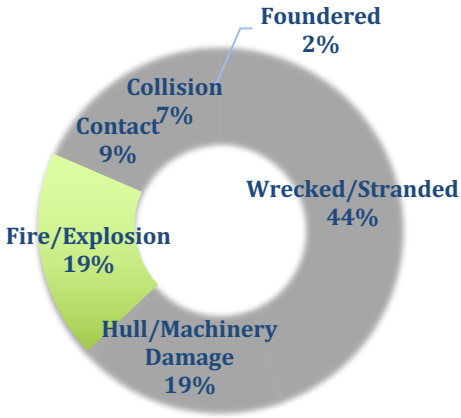
[ferries] which would bring the total of human casualties to 10, and the number missing to 28.) This means that in 64 years, at most 38 people have lost their lives as a result of a passenger vessel incident in the Arctic. But there have been incidents requiring the rescue of at least 1243 passengers and crew, all of whom were rescued safely. The three incidents involving the highest number of passenger rescues all have occurred between 55 and 60 degrees latitude, and were described by the following:

"Cruise ship caught fire 120 miles away from land in heavy weather. The vessel was still burning when it sank. All 300 passengers and 200 crew were rescued safely."

"Ship sustained punctures to forward fuel/water tanks. 122 passengers and 31 crew were rescued by fishing vessels."

"Stranded on the rocks when leaving Hubbard Glacier, divers' inspection revealed 10 ft long, 5 inch wide gash in ballast tank. No injuries reported to the 2,000 passengers, temporary repairs made and ship continued after Coast Guard inspection."

Most Common Incidents Cruise Ships



The most common incident for cruise ships is running aground

Historically, the greatest number of cruise ship incidents have been caused by running aground, followed by an equal number of incidents involving hull/machinery damage and fire/explosions. Of the incidents involving large numbers of passenger rescues, 8 were from running aground, and 2 involved fires. Passenger vessels which were repeat offenders tended to be ferries, with only 2 cruise ships being involved in more than one incident.

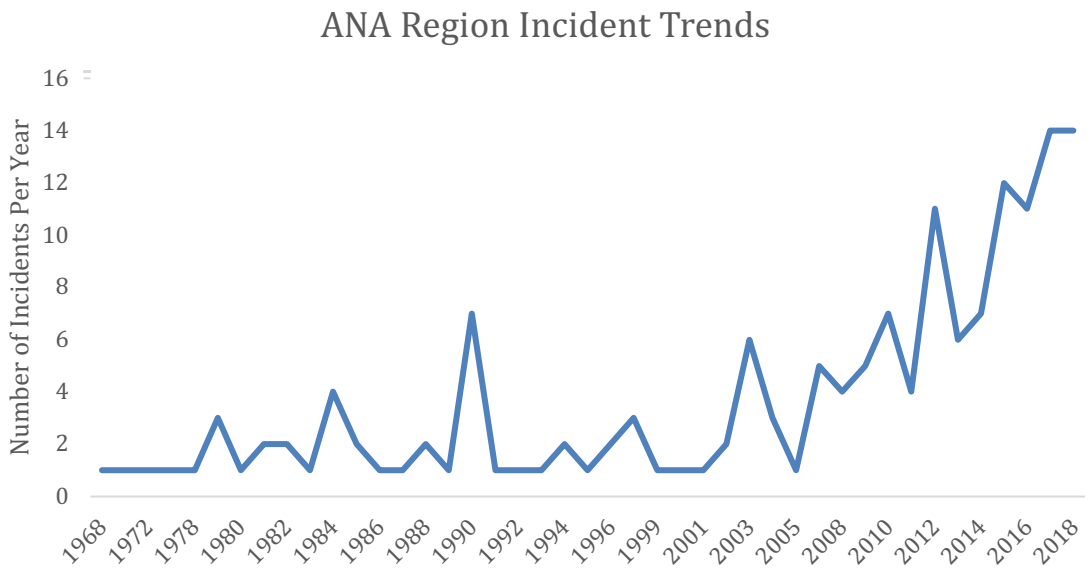
Arctic Incidents at a Glance:

Number of Incidents Extracted: 1822
Average year of build 1979
Average Gross tonnage: 7836
Average Deadweight: 10535

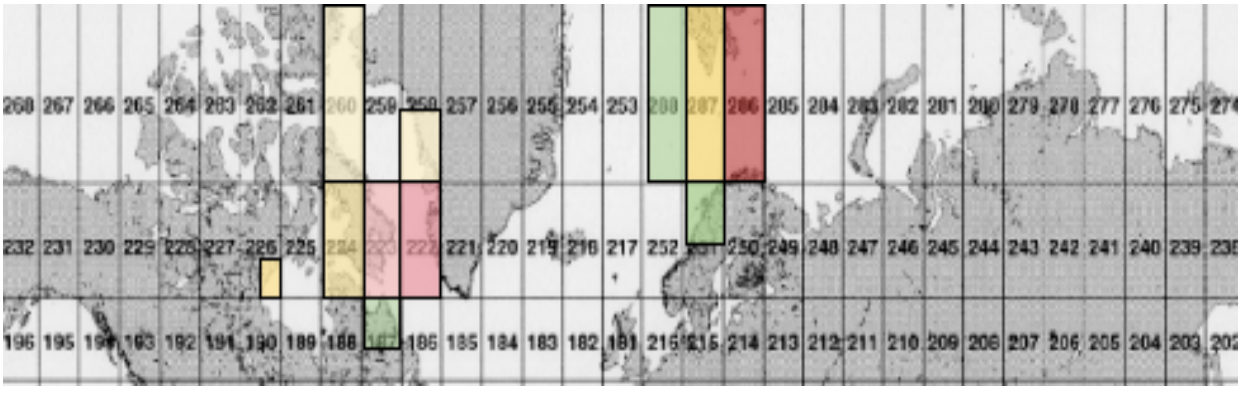
Number Killed: 282
Number Missing: 486
Pollution Noted (# of incidents): 98 (5%)

Arctic North Atlantic Region

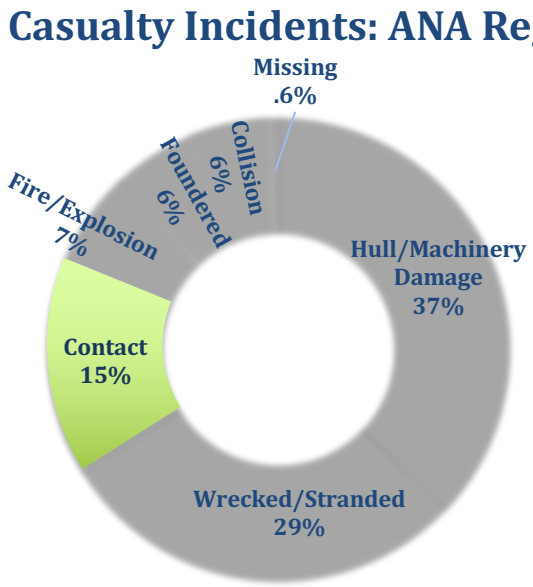
Since the main focus of the current analysis was to examine the Arctic North Atlantic specifically, we then wanted to then examine the trends and incidents occurring specifically in this region. Utilizing the Marsden Grid References in the all Arctic incident database (where available), we were able to clearly identify 160 incidents as occurring in the ANA region. We then conducted similar analyses as those done for the all incidents database. (It should be noted that the analyses reported earlier are inclusive of these 160 incidents that occurred in the ANA region.) In this region, as in the Arctic as a whole, marine incidents are on the rise.



Utilizing the Marsden Grid (below), we see the that the highest number of incidents occurred around the Murmansk area and Northern Norway, followed by the Baffin Bay region of southwestern Greenland.

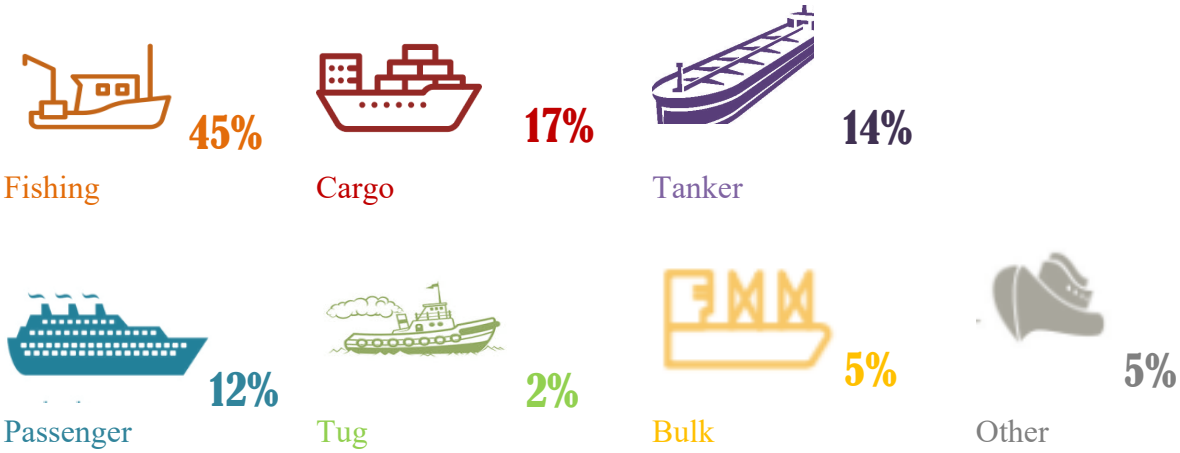


Similar to the entire Arctic region, the most common casualty type in the ANA Region was Hull/Machinery Damage followed by Wrecked/Stranded, with the least likely incident after missing vessels being due to collisions and foundering.

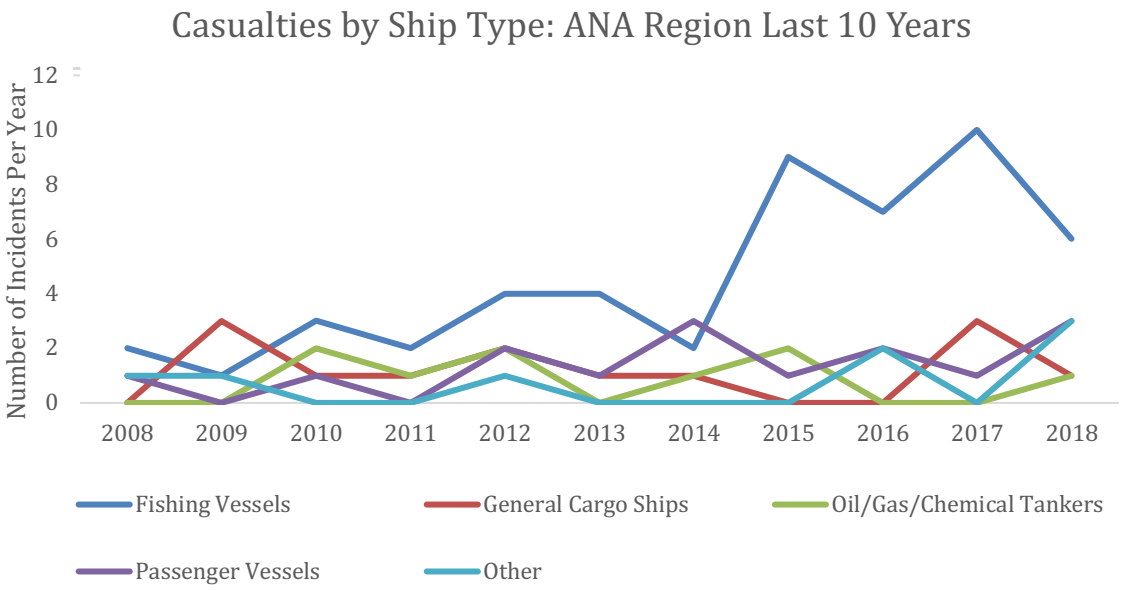


The most common type of casualty was damage to hull or machinery, followed by wrecked or stranded.

Incident Ship Type in the ANA Region



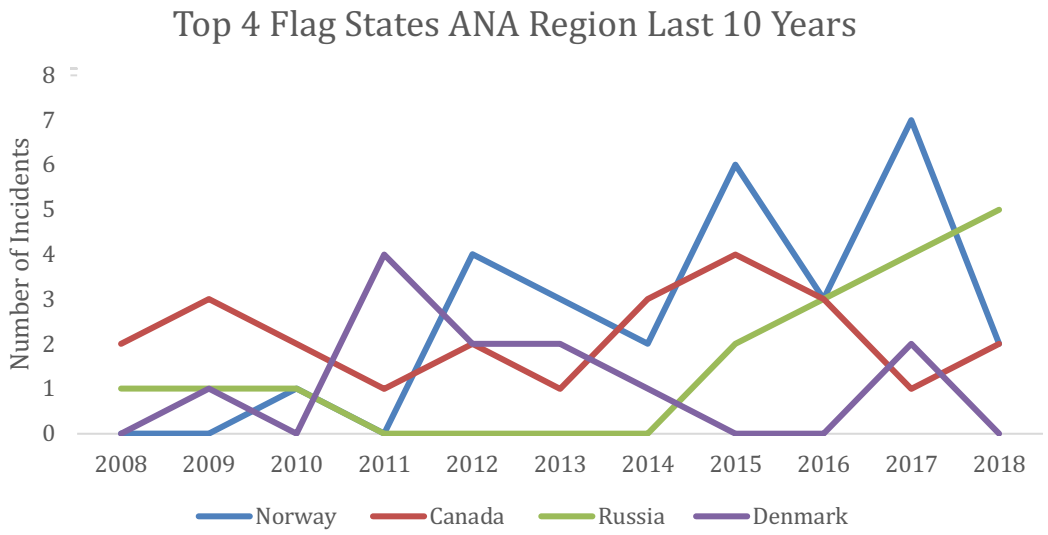
Also mirroring the trends in the entire Arctic region, fishing vessels accounted for the largest number of incidents in the ANA region (representing almost half of all incidents). However in this region, the percentage of incidents involving oil/gas/chemical tankers is slightly higher (14% vs 11%), as is the number of incidents involving passenger vessels, as compared to the entire Arctic region (12% vs 7%). Examining the trends over the last 10 years (below), we see that fishing vessel incidents have increased dramatically, beginning in 2014. There was also an increase in 2014 in passenger vessel incidents, which was not sustained in the following years.



ANA Region Incident Flag States (1955-2019)

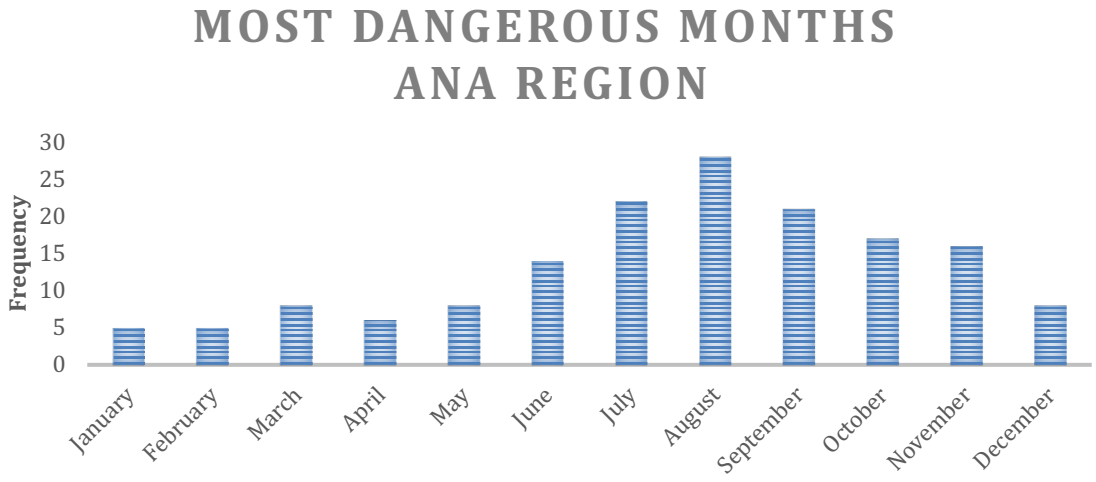
Perhaps the biggest difference between the data for the ANA Region as compared to the larger Arctic as a whole corresponds to the flag states of ships with the highest number of incidents. In this region, the majority of incidents occurred to Norwegian flagged vessels, followed by Canadian and Russian.

We also see a greater amount of variability over the years in this region as to the number of incidents per flag state. Of note is the steady increase in incidents for Russian flagged vessels since 2014, and the alternating years of incidents for Norwegian flagged vessels.

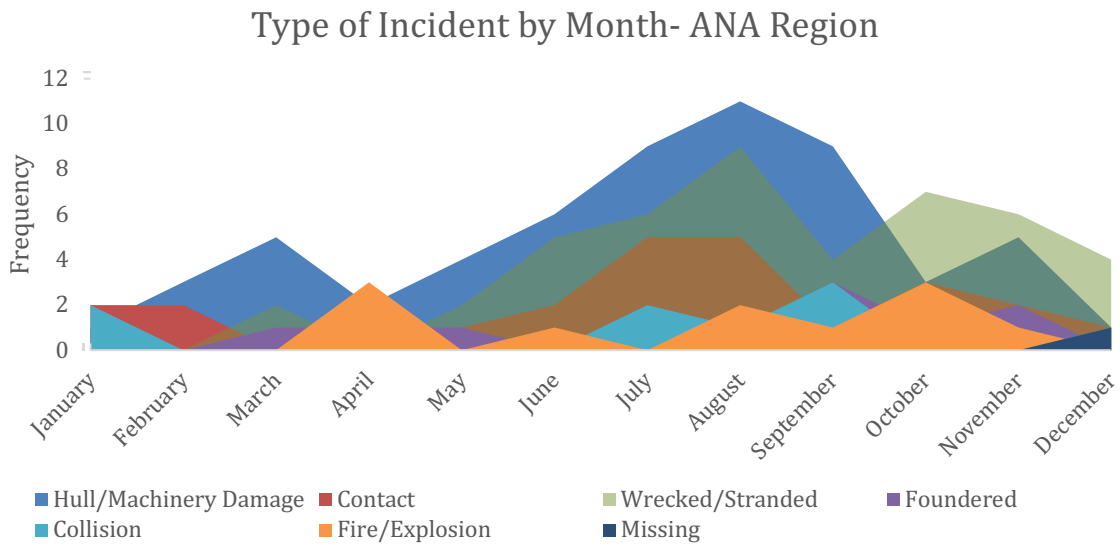


A Timeline of Incidents: ANA Region (1955-2019)

The highest number of incidents in the Arctic North Atlantic region occur in the summer months of August and July, followed closely by September, corresponding likely to the reduced extent of sea ice. Whereas this is similar to the findings for the Arctic region as a whole which saw incidents spiking in August, the two areas do not share similar incident rates in the month of February. Here, February and January are tied as the safest months of the year.

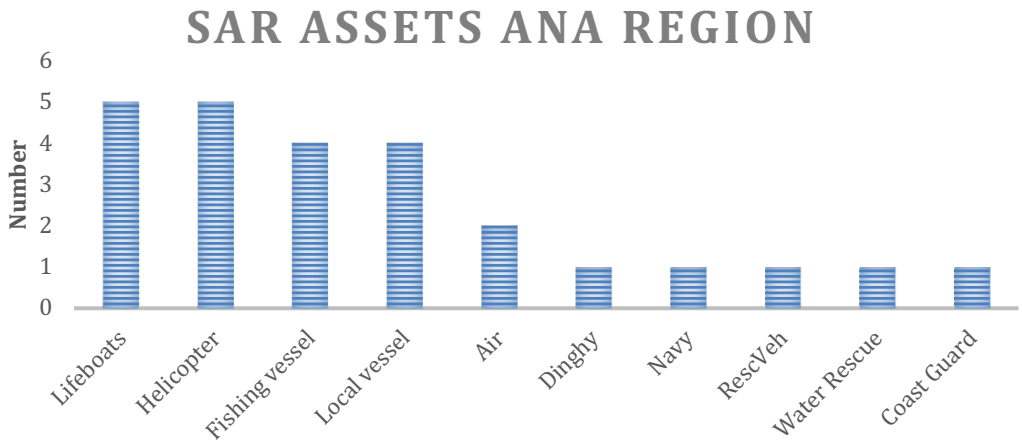


Examing the type of incident by month, also presents a less clear cut picture as to the reason for the higher numbers of incients in August than for the Arctic as a whole. The highest number of incidents remain due to Hull/Machinery damage, follwed by Wrecked/Stranded and Contact. Even though summer months are often designated as "ice free," we see that there is still a hightened chance of hull and machinery damage during this time.



Search and Rescue ANA Region

Only 19 incidents in the ANA Region database had sufficient information to examine SAR resources. Of those requiring rescue, 4 crew members died, none were listed as missing, and 44 required additional medical attention. At least 163 were rescued safely (again with the caveat that this number is likely higher due to the entries stating that 'all crew' were rescued safely). In this region, similar to the Arctic as a whole, the most common rescue assets were other vessels of opportunity (n=9), followed closely by helicopters and lifeboats (n=5).



ANA Region Incidents at a Glance:

Number of Incidents Extracted: 160
Average year of build 1985
Average Gross tonnage:5942
Average Deadweight: 6613

Number Killed: 6
Number Missing: 13
Pollution Noted (# of incidents): 9 (6%)

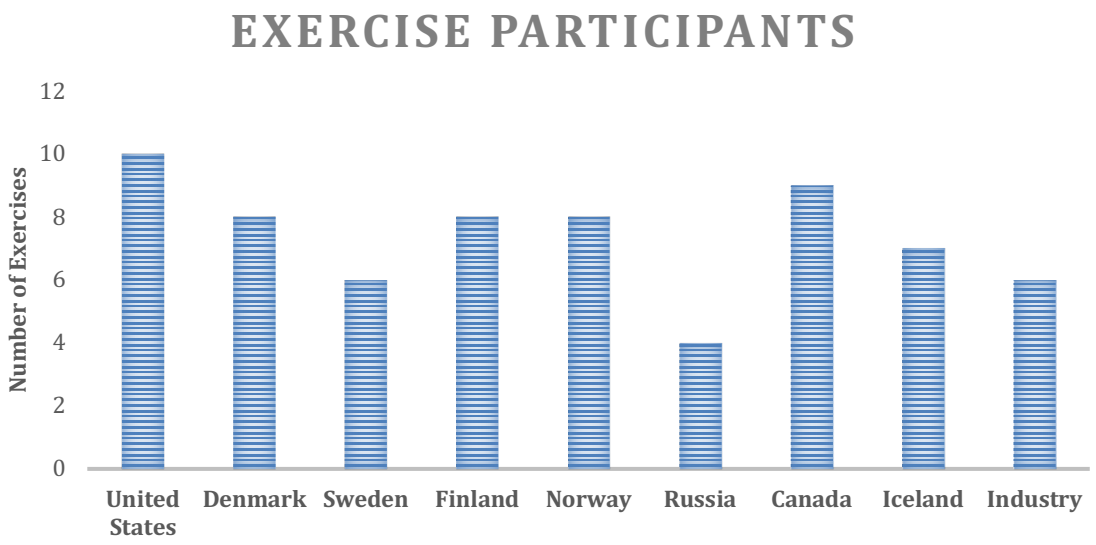
Live and Table Top Exercises

To see where state and industry leaders are both placing their priorities and to evaluate readiness, and to get an idea of what they believe to be the most likely scenarios in the future, we also looked at a number of exercises that have been conducted both live and notionally in the region. From our research, there have been three types of operations that have been conducted to test readiness by actors in the region; search and rescue, environmental response, and military operations. We hoped to study the after-action reports of these exercises to understand common themes of gaps in preparedness. Unfortunately, of the 25 exercises we identified, we were only able to access 11 after action reports from the source of the exercises. The nature of many of the response organizations directly reporting to or affiliated with defense departments led many of them to be classified documents.

Although in many cases there were newspaper articles or some other secondhand accounts written about the events that were classified, we thought that narrowing our analysis to direct accounts of the exercises would give the best sample. Despite this small sample, we still believe themes can emerge in terms of readiness, cooperation and the way actors are thinking about potential disasters in the region.

Participants

Of the 11 cooperative exercises we were able to access the reports for, all the Arctic Council member states were represented in at least one of the exercises. Our selected exercises had different hosting organizations including the Association of Expeditionary Cruise Operators, the Arctic Coast Guard Forum, or bilateral operations organized by the respective countries.

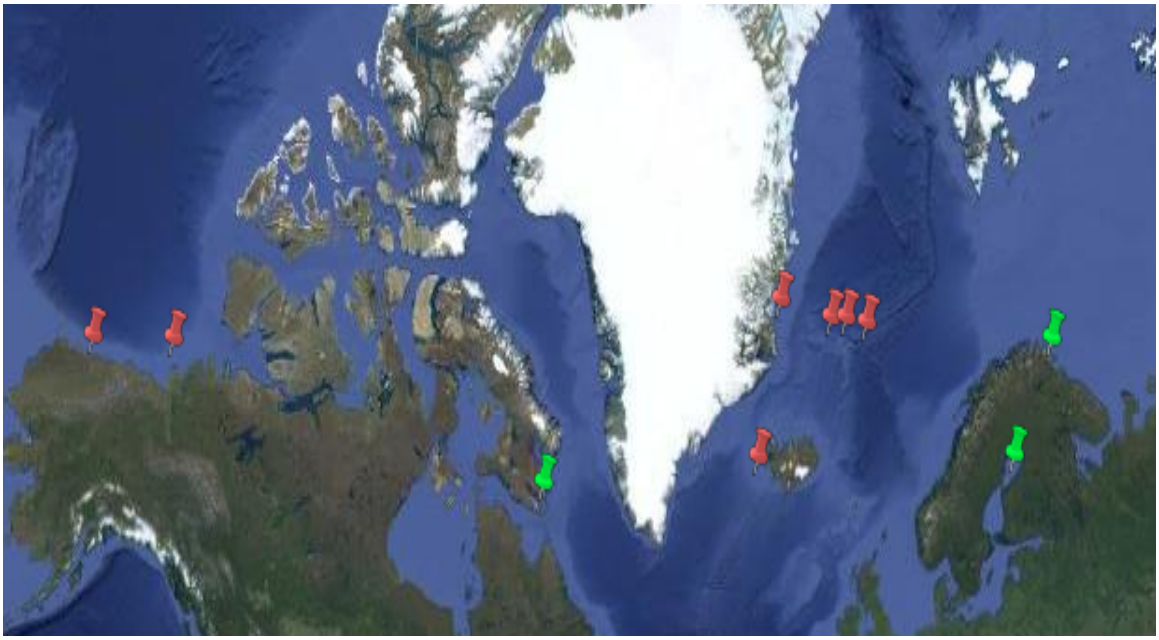


**(Not pictured are Italy, Singapore and the World Wildlife Fund who each participated in one of our exercise samples)*

Exercises

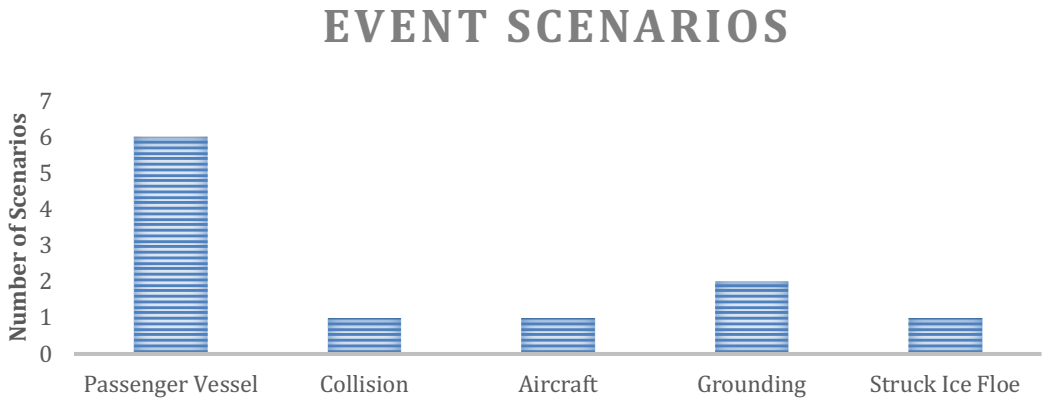
Exercise Name *Exercise Date* Exercise Type *Lead Country/Agency*	**Arctic Shield** *July 2015* Live SAR Exercise *USCG*
SarEx Greenland Sea *September 2013* Live SAR Exercise *Denmark JACO*	**Arctic Zephyr** *September 2015* TTX SAR Exercise *USCG/USDOD*
Arctic Guardian 2017 *September 2017* Live SAR Exercise *ACGF/Finnish Border Guard*	**Northwest Passage** *August 2016* TTX SAR Exercise *USCG*
AECO 2016 *April 2016* TTX SAR Exercise *AECO*	**AECO 2017** *April 2017* TTX SAR Exercise *AECO*
AECO 2018 *June 2018* TTX SAR Exercise *AECO*	**Kalajoki 2013** *March 2013* Live Environmental Response *Finland Environmental Institute*
MOSPA 2016 *May 2016* TTX Environmental Response *EPPR/United States*	**MOSPA 2014** *May-June 2014* TTX Environmental Response *EPPR/Canada*

As is evident from the above table, were able to access a variety of exercises, and extract information regarding the type of response and whether it was a live exercise or notional, involved a search and rescue and/or environmental response operation. A map of the locations of these joint exercises can be found below.



Common Themes

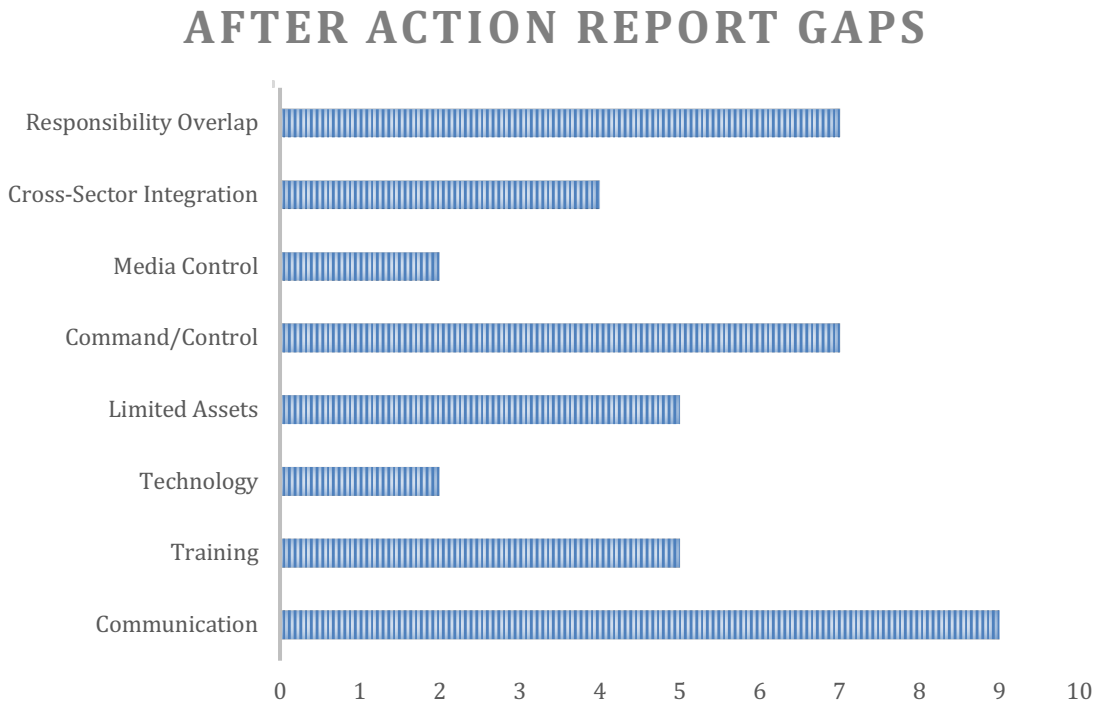
Across the eleven exercises, we found that the most preparation was being done for a cruise ship disaster.



Of the eight SAR exercises we were able to access, six were focused on a hypothetical cruise ship accident. This suggests a consensus as to the need to prepare most for this particular kind of incident as it represents the worst-case scenario in the minds of rescue operators in the Arctic.

Gaps in Capabilities

There were a number of gaps identified in the after-action reports. The most common gaps identified involved communication and coordination issues, in the form of responsibility overlap and command/control. Each of these will be discussed in more detail below.



Responsibility Overlap
One common finding from our analysis of available reports was a concern with overlapping responsibilities. Overlapping responsibilities occur at high levels which can cause confusion and redundancy for training. (High level being defined as the national and multi-national level.) Both in tabletop exercises and live exercises, participants often expressed a desire to have additional clarity in terms of their areas of responsibility for operations.

Cross-Sector Integration
Another concern that emerged from these reports was a lack of integration with non-government actors, specifically environmental groups and maritime industry. Environmental groups were mentioned so that they can share their expertise and concerns for events that might occur in the region. The maritime industry was mentioned for some of the same reasons, but also in terms of the assets that they may contribute to emergency response. A common conclusion was that these assets would improve effectiveness in real-life scenarios, and as such should be included in the planning and practice of emergency response. The fact that this concern was mentioned, even though over half of the exercises examined included some industry or environmental non-government actors, is notable.

Media Control
A few reports mentioned concerns regarding media communications, fearing that the media may compound on-scene panic and disorder. The importance of ensuring that correct information is being passed to passengers and media allows for other parts of a response operation to go as smoothly as possible.

Command/Control
Another common concern from these reports was the confusions in command and control elements of the exercises, and how it could translate to impacts in real-life events. In some cases, especially in higher levels, there was confusion as to who was in charge of a particular area or stage of an operation. Role distinctions such as *on-scene coordinator* versus *search and rescue mission coordinator* appeared to created points of friction for operators during the exercise.

Geographical distinctions created confusion in terms of command and control as well. Who took over when operations moved to land or across a particular boundary, for example, was mentioned as a problem in some of the exercises.

Limited Assets
Five of the reports made mention of the need to overcome the limitation of assets when responding to incidents in the region. Assets could be defined as response vessels, helicopters, fixed wing aircraft and on-shore resources like medical care facilities and personnel. The reports do not make normative statements in terms of requesting more, but more to facilitate an awareness to those doing business or preparing for response, that there are fewer resources in the area for emergency response. Especially in the table-top exercises that involved industry, the role that private vessels and their associated assets would have in responding to emergencies was highlighted.

Technology
Technology gaps unique to the nature of our region were also brought up by a few of the reports we reviewed. In particular, the lack of satellite imagery, communication and cold-water specific life rafts were brought up as concerns. Search and rescue agencies noted the lack of infrastructure to allow for a clear picture of weather conditions, and the inadequacy of life rafts during the exercises. Reports showed clear deficiencies concerning the life rafts, and difficulties in the ability to extract survivors from them.

Training
Another common theme that emerged as gaps in the reports involved training of both response agencies and crews onboard vessels in the maritime industry. Recommendations noted that responding agencies should either increase the number of the exercises, or cross-train with other agencies and countries, to hone their own skills and get perspectives of partners.

For crews on private vessels, gaps were identified in the lack of clarity in the role that crews would play in an emergency event. The reports indicated that industry did not effectively delegate responsibility to different crewmembers on their vessels, which could aid in a timely and organized evacuation in the event of a disaster.

Communication
The most common gap identified in the after-action reports was issues with communication. Communication, for our purposes, is widely defined to cover a number of issues. For example, reports showed a lack of common systems for communications, points of contact that had outdated names and numbers, or something as basic as a common language for those operating radios. Live exercises, table-top, search and rescue and environmental response all mentioned some deficit in communication in their practice situations.

Many of these gaps seem to overlap, but in many ways appear to reflect the same gaps that we identified in our interview responses. One of the key takeaways may be that the best way to fill these gaps is to continue to exercise so that they can be identified and solved.

Interview Data

Participants

We reached out via email to 51 individuals requesting interviews for the current project. These individuals were inclusive of a number of Arctic shipping stakeholders, including shipping companies, insurance company representatives, environmental groups, military, coast guards, and rescue operation centers. A total of 28 people agreed to be interviewed for the study, most of which were conducted by phone or Skype. Three interview participants preferred to respond to the questions in writing. Our interview subjects included individuals from all Arctic states represented in the Arctic and North Atlantic region (see table below provides a general overview of the participant pool). Interviews took, on average, about an hour and we asked the same questions of each respondent to ensure comparability across individuals (see Appendix A for the list of questions). Confidentiality was assured to our participants to encourage them to share openly. Interview responses were transcribed, and then coded utilizing NVivo software.



Academics/Think Tanks

6



Environmental Group

3



Military/Coast Guard/Rescue Centers

7



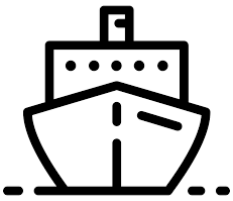
Shipping Insurance

3



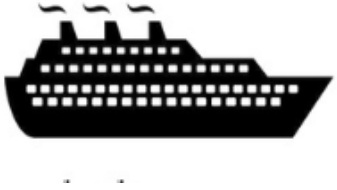
Government

5



Shipping Company

2



Cruise Ship Industry

2



Arctic Council Working Group

2*

** Two of our interviewees, although representatives of one of the industries noted above, were also members of an Arctic Council working group.*

As is clear from the table above, one of the shortcomings of our study was our inability to interview spokespersons from Arctic Indigenous groups, nor any representatives from Classification societies. We reached out to multiple members of each group but were ultimately limited to who responded to our interview requests. It should be noted however, that including perspectives from both of these groups would have provided important perspectives regarding industrial development in the region.

In our interview questions, we defined our region of interest as the area between the eastern coast of Canada and Norway. All of our interview participants rated their familiarity with the Arctic North Atlantic region as very good to intimate. It should be noted that although we asked interview participants to constrain their responses to the Arctic North Atlantic region, most of them had difficulty responding with regard to just this area and expanded their focus and responses to include Russia and the Northern Sea Route as well (n=17). We believe the reasons for this may be twofold: It appears that many of our respondents felt that the majority of the development was happening in the Russian Arctic as opposed to the North Atlantic region. Second, a number of participants responded that the issues that would be faced in the lower portions of the North Atlantic region would be similar to issues that were present in the Mediterranean or "*the rest of the world,*" given the lack of sea ice. Therefore, although our goal was to define our area of interest to only the Arctic North Atlantic, many times our respondents included a more expanded view in their responses.

Is Arctic Shipping Increasing?

Whereas nearly everyone interviewed felt that shipping in the Arctic was increasing, there was some disagreement as to the extent. There was wide consensus that this expansion was happening only with regard to destinational traffic, and even then, isolated only to certain types of ships (most of which did not involve bulk container shipping). There was also general consensus that the increases did not match levels that were originally predicted. *"The way it's reported, as massive, massive increases in traffic in the northern sea route, the traffic increases by 25%. Well, 25% of 12 ships is how many ships?"*

Our interviewees were likely to cite misconceptions surrounding the use of the term "ice-free," and consensus emerged regarding the extent to which the Arctic is, and will remain for the foreseeable future, a hostile environment for shipping. Many of our participants noted that despite the receding ice, rapid changes in weather and ice conditions continue to make the Arctic too dangerous and unreliable for large scale, transit cargo shipping (n=5). *"I think the hyperbole around climate change is giving people the wrong impression about how easy it's going to be to operate in those environments."*

Both shipping company representatives were in agreement with this assessment, with one stating, *"Transit shipping is not going to happen in the Northern Sea Route anytime soon, and the Northwest passage will always be useless for commercial shipping transits. Economically, it doesn't make sense to transit through Arctic when you consider the cost per unit cargo, it's simply too expensive compared to the alternatives."*

These costs could further increase with regulations under consideration, namely a ban on the use of heavy fuel oil (HFO) in the Arctic. *"An HFO ban would require engine modifications to fuel pump and filter, and light oil is 2-3 times [more expensive], so it would be a big expense, which then makes shipping in the Arctic even less economically feasible."* It would appear then our respondents were not in agreement with the predictions that receding ice will result in drastically higher numbers of commercial ships transiting through the region. One academic summed up the general consensus, *"I think [shipping is] going to increase, but the level I think is anybody's guess in the next 15*

years, with respect to the price of oil and how much it costs to move ships around in southern waters, which are so much easier than Arctic waters."

Despite the fact that our respondents discounted widespread increases in commercial cargo ship traffic in the Arctic, there was consensus as to more specific concerns surrounding untrained crew, and the increase in particular types of ships. Inexperienced crew operating in the region was mentioned by a large number of our interviewees (n=7). A common response was that if there were to be an increase in numbers of those currently and historically operating in the area (e.g., resupply ships), it would be of less concern than an increase in ships from new, less established industries or crew. Four respondents explicitly mentioned serious concerns about ice inexperienced crew sailing in polar waters.

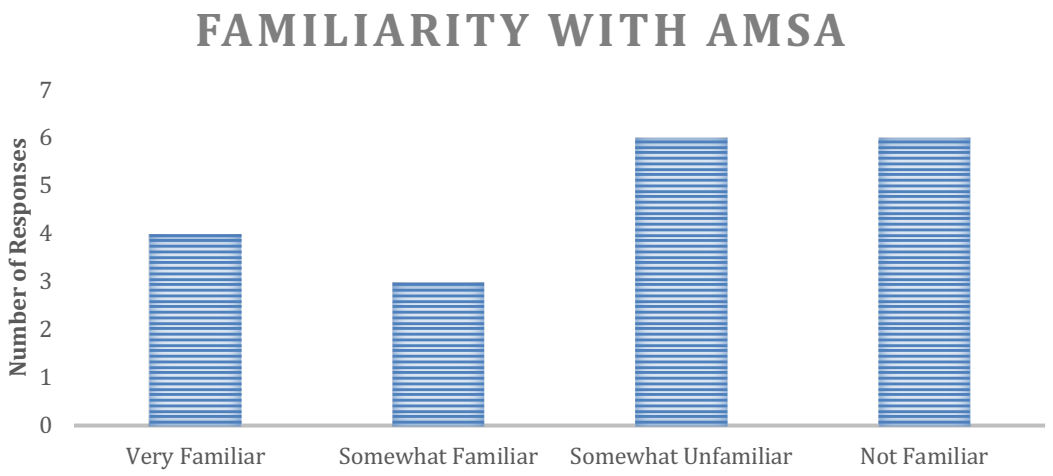
"You will not find a requirement for actually having been trained in Arctic regions. For me that's not, you cannot imagine cold if you haven't experienced it, and you cannot imagine how remote the area is if you've only, been sailing in the Indonesian waters."

"You [can] get a certificate and then you can be a captain on a tanker or a passenger ship, be a chief officer, you can take that ship anywhere in polar regions under the polar code, but you've never seen a stick of ice in your life and you've never had to maneuver around it…and they have no experience in ice."

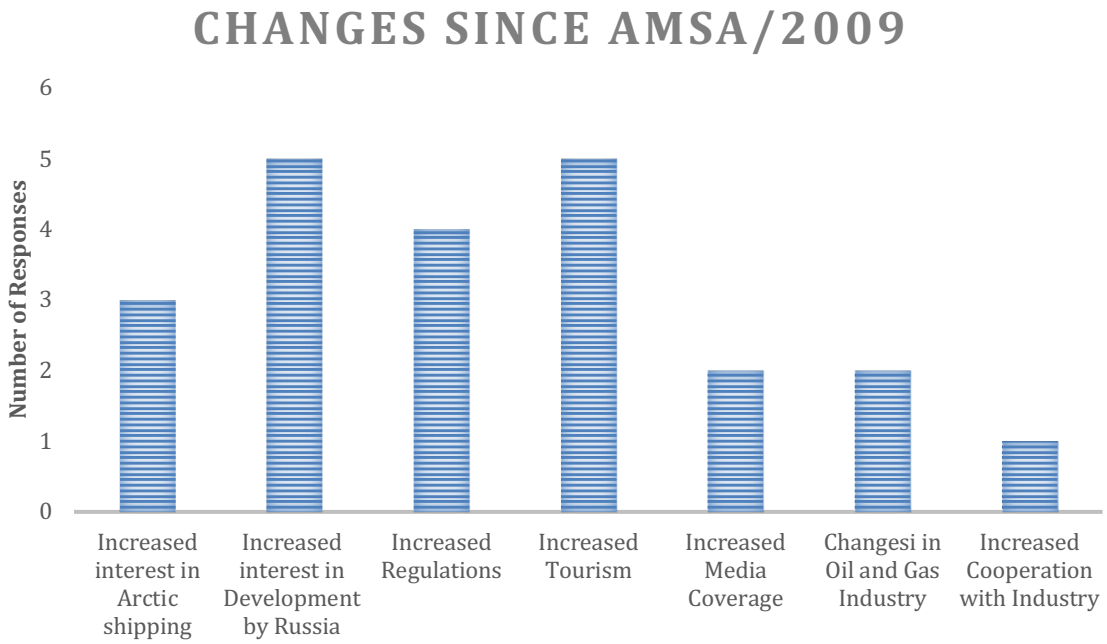
Therefore, on the whole, there seems to be the consensus that the answer to whether shipping is increasing in the Arctic and North Atlantic region is yes, but not to the extent that had been predicted, and depending on industry, is not necessarily a large cause for concern. Predictions that receding ice would open up the area to massive increases in containerized or cargo shipping has not been borne out by the data and, at least according to our interviewees, is unlikely to change anytime in the near future. Increases in established areas of shipping were considered to be less of a problem by our respondents, compared to the growth in the number of inexperienced individuals operating in the area stemming from increases to certain types of industry.

Changes in Industry

We asked respondents about the changes that they've seen in the ANA region since the publication of the 2009 AMSA report. A large number of our interviewees were surprisingly unfamiliar with the report (see Figure below).



Even among those familiar with AMSA, a few were hesitant to attribute changes they've seen in the region directly to the report. For this reason, and because of the low level of familiarity in our sample, we asked any who expressed discomfort to simply respond about changes they've seen in the industry since the time of the report's publication (2009).



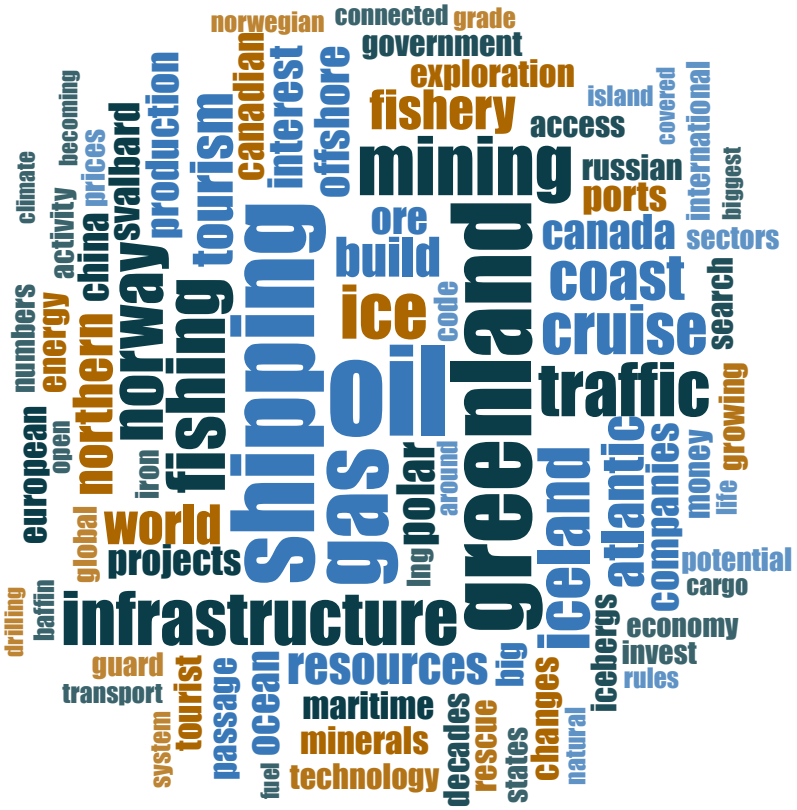
Among those familiar with the report, the general consensus was that the AMSA raised awareness of the potential for increased Arctic shipping. A number of respondents also felt that the AMSA paved the way to greater regulatory oversight (including the eventual adoption of the Polar Code), brought increased awareness to the need to protect the Arctic environment, and enhanced Arctic marine safety.

Developing Industries

We then asked our interviewees to describe the current industrial development in the region, along with projecting which industries would be present in the next 15-20 years. Responses showed a great deal of consistency and tended to fall into 4 main industrial areas:

- Oil and gas, including offshore development
- Mining
- Fishing,
- Cruise/tourism

Again, it should be noted that commercial containerized shipping was not present in these projections, in fact, only 2 respondents noted that it was likely to increase in the next 15-20 years. The figure below presents the most commonly mentioned keywords included in interviewee's responses.



Mining

Almost half our interviewees (n=15) mentioned the ongoing mineral extraction in Canada, Norway and Greenland as both sources of current shipping, and projected into the next 15-20 years. Citing the rich mineral resources that have been found in the Arctic region, our respondents felt that this would be a continued and reliable source of Arctic shipping, especially as more minerals are discovered.

Energy Production
Interviewees were also likely to mention oil and gas development in Norway and Russia as sources of current and future shipping. Although Canada and Greenland have recently prohibited new oil and gas exploration, our interviewees felt that if the price of oil were to increase to over $100USD per barrel, both oil and gas exploration and increased attention to the viability of shipping through the arctic regions would increase. According to one industry representative *".. oil and gas exploration ... also will be in the future, there will be a need for energy in the world and as long as there are no alternatives, this will be following the other developments in the Arctic."*

The price of oil would also influence the extent to which North Atlantic offshore development would continue into the future, with the exception of Russian LNG exports. There was near unanimity in the belief that Russia would continue to expand its LNG production in the future, and since a main route for this LNG is through Norwegian waters, it can be expected that this type of traffic will expand in the future as well.

However, oil and gas are not the only sources of energy production being considered in the area. 5 people mentioned the possibility of coastal alternative energy development stemming from water and ocean energy production. Others forecasted technological advances in fuel might increase levels of shipping in the Arctic, and may even bring about greater commercial shipping in the future. *"So I would say that [shipping] will evolve through technology, maybe LNG powered ships in the future."*

Cruise Tourism
Of all of the industry development discussed by our interviewees, the one that received the most attention was the present and future increase in cruise vessel tourism all over the Arctic North Atlantic region. Respondents expected not only a greater number of ships, but also an increase in the size of ships that would be coming. Three respondents noted that there is already a number of new ships in production, which are specifically being built for Arctic cruise expeditions. There was also some consensus (n=5) projecting that the expedition cruise ships would continue to seek out more and more remote areas.

Once again, however, the projection in expansion was qualified by global economic considerations by our participants, who pointed to the high price of these types of cruises (n=3). Four respondents expressed additional concerns regarding the impact of cruise tourism on local populations. One participant in particular brought up the recent weekend moratorium placed on tourist visits[67] by the Faroe Islands, as an example of what could happen if the increase in cruise traffic results in overtourism. She also noted that "*when there's 27 vessels coming in, probably not all biggest, but even if you take just a thousand people more and everybody's really interested in learning about the local people without really having time to do it in a proper way, they are really intruders more or less."* (LH)

Fishing
Because fish are moving farther north as sea temperatures rise, a large number of our interviewees also mentioned the fishing industry as one of the major areas of both recent and future industrial development. This development, including both fishing but also the transshipment of fish to other areas, is expected to increase in the entire ANA Region.

Because of this increase, it was not uncommon for some respondents to mention the fact that fishing vessels are outside of the polar code regulations (n=3). *"You know fishing vessels as a general rule often aren't subject to the same safety standards as other vessels are, there's always a bit of concern there, and they tend to they follow the fish, and that often leads them to stray. There's more risk of a fishing vessel I think getting caught in trouble than a larger tanker."* Another noted, "*the fishing industry has to find more remote locations if they want to find fish*."

Of those who mentioned it, our interview participants had generally positive things to say about the fishing moratorium in the high arctic, noting the importance of understanding how climate change will influence fish stock before allowing fishing in the High North. There were some concerns noted however as to what will happen when that moratorium expires, especially with regard to non-Arctic states like China, Japan and Korea, who may seek to expand their fishing activities in to the area. *"The fisheries in the North Atlantic are well controlled and managed, I think. But how much pressure will be brought to bear by European fishing companies, and others from China?"*

A few of our respondents (n=2) also noted that this migration of fish might be a source of conflict in the area in the future. *"The closest that NATO allies came to war with each other was over fishing."* A few individuals also expressed concerns over the lack of infrastructure and ability to police illegal fishing, but also included concerns about how ownership of fish stock may pose problems in the future.

"[Right now] the Norwegians and the Russians cooperate really well on fish. Well, the crabs are starting to move east into Russian waters. And the Norwegians are claiming they're still our crabs, and there's a lot of tension over who gets the crabs."

"There will be an issue as to some states encroaching on our fishing waters, and the US is currently incapable of patrolling that area. Right now we have an inability to exercise sovereignty over our waters."

In sum, the general consensus seems to be that there will continue to be increases in Arctic shipping, but they will be constrained to four general areas. Two of these areas seemed to present more of a concern than the others in our interview sample, cruise vessels and fishing, perhaps due to the fears that these industries would continue to seek out more and more remote locations. Concerns were also raised as to the potential increase of inexperienced operators in the area that may accompany expansion of these industries. As one government participant noted, *"but you know, these ships kind of fall below the regulatory threshold in many cases, and many more want to go off the path and, do stuff that's less regimented and I guess a lot of the smaller ships whether it be pleasure craft operators, adventurers, even fishing vessels to some extent just don't have the history of the experience operating in the Arctic, compared to a lot of the fleet, the cargo ships, tankers etc. that have been operating in region for decades on end."*

When Is the Increase Significant?

We then asked our interview participants when the increase in shipping would be significant or would signal a real change in levels of Arctic shipping. In this question, we saw quite a bit of variability in the manner in which people responded, as captured in the Table below. The majority of respondents felt that numerical estimates were less useful than focusing more on types of events or ship characteristics that would signal a significant increase.

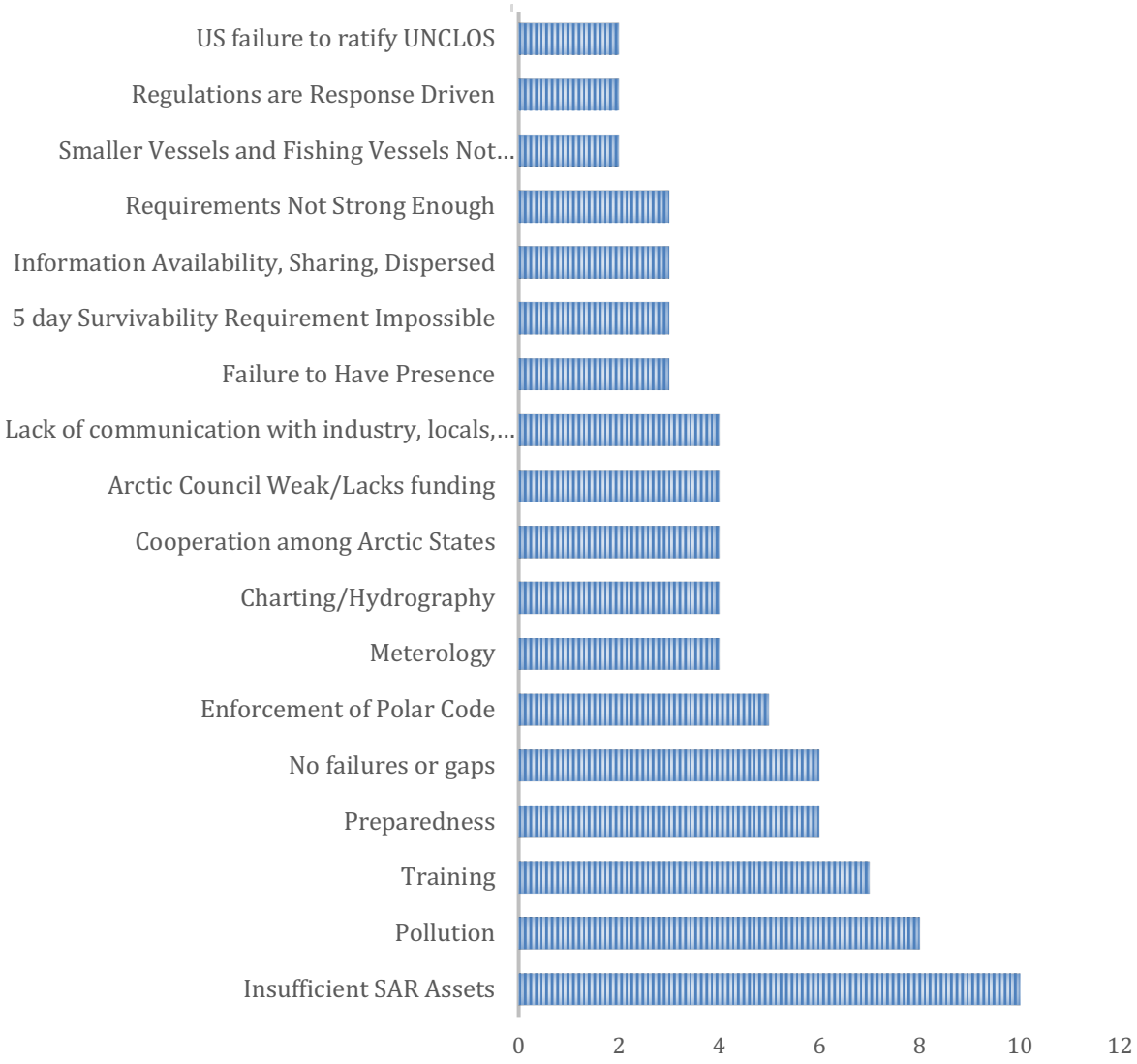
Q: What would be a significant increase?...	***Examples***	#
Provided numerical estimates	◊ 20-30% over 3-5 years ◊ 30-50 through NSR, 15-20 NWP ◊ Couple thousand more	3
"Not the number but the type"	◊ Large influx of inexperienced crew ◊ Offshore oil leases resume (2) ◊ Big cruise ships in the area	4
"When it exceeds local tolerance"	◊ Overtourism ◊ Only a couple more from what is already impacting small communities ◊ When it impacts economic development of local communities	4
Provided time estimate	◊ Now ◊ Sooner rather than later ◊ In a 30 year horizon	3
Provided an event	◊ When fish move north and industry follows ◊ When industry finds it profitable to operate there ◊ When ice melts ◊ When there's a significant increase in ice classed vessels (2) ◊ When regular levels of cargo shipping ◊ When there's regular transits through NWP	7
"When it exceeds SAR Capacity"		6
"Not the number but the size"		3
"When ships start going into remote areas"		2
"I don't think the changes are significant"		3

Critical Gaps and Failures in Governance Structure

There was also a fair amount of variability in responses to our question regarding critical gaps, or major points of failure in the governance structure in the Arctic. The most common responses referenced a greater need for search and rescue assets (e.g., helicopters, infrastructure, individuals with sufficient SAR training, etc.), a need for stricter requirements regarding environmental protection (e.g., pollution, heavy fuel oil, grey water, research on oil in icy areas), and/or a need for stricter regulations as far as crew training for those operating in the area.

However, six interviewees believed that there were no gaps or major points of failure in the current governance structure, with an additional 3 respondents expressing the belief that "*Regulations aren't the best way to bring about changes in this area,*" or "*Regulations aren't the only way*." Therefore, 9 out of our 28 interview participants were satisfied with the current regulatory status quo. One participant representing the shipping industry stated, "*There hasn't been, knock wood [sic], a serious accident in the Canadian or American Arctic that I can recall in 20 years? And that's a testament to the fact that there are no critical gaps or major points of failure in the current governance structure."* Six respondents cited the two recent agreements coming from the Arctic Council, SAR and MOSPA, as evidence that the current governance structures were operating successfully.

On the other hand, those in favor of strengthening either the regulations, or the regulatory agencies (e.g., IMO, Arctic Council) were most likely to recommend either greater enforcement authority, or stronger regulations. Three of our respondents felt that the current regulations in the form of the Polar Code were too "watered down," or were not expansive enough in their coverage (e.g., polar code needs to be applied to smaller vessels and fishing boats, training requirements need to be stricter, or all aspects need to be mandatory). Three interview participants were also concerned about the lack of funding for the Arctic Council, with one stating that it created a regulatory body with "*no teeth."* Some expressed similar concerns with the IMO stating that it was slow, limited or not focused enough on unique features of individual states.



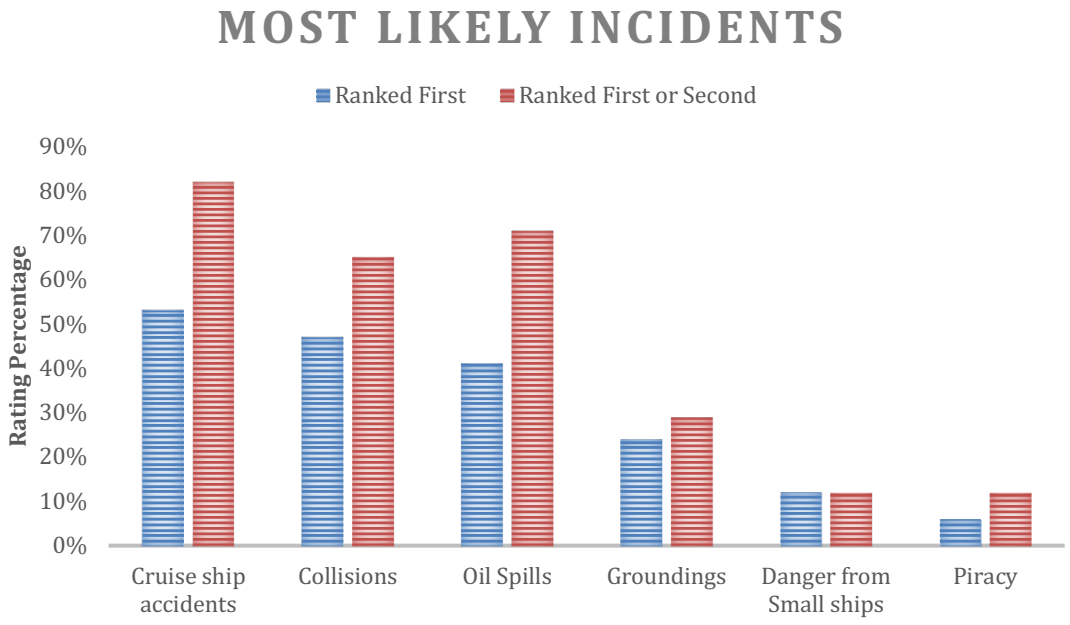
Which Incidents are Most Likely to Occur in the ANA Region?

Because the main goal of our study was to forecast the likely ship traffic and incidents which are likely to occur in the ANA region in the coming years, we asked interviewees to rank the likelihood of three events we presented to them, including seaborne disasters (e.g., collisions or groundings), catastrophic incidents (e.g., including cruise ship accident or oil spill), and soft security threats (e.g., human trafficking, piracy, transport of illegal goods, or failure to adhere to polar code). There was a lot of variability in responses to this question as well, indicating a lack of consensus among our experts as to which incidents would be most likely. Examining the nature of their responses, it was also evident that our interview participants had difficulty separating out the issue of likelihood from consequence. Understanding that *risk = probability x consequence*, many preferred

to focus more on the consequence portion of the equation, rather than the most likely event from a probabilistic viewpoint. In fact, a few of our respondents told us explicitly that looking at risk **of** an event was less useful than looking at the risks stemming **from** a particular event.

There was near universal agreement that a cruise ship disaster would be the biggest threat to the area, thereby posing the greatest risk. There were also very high levels of consensus that the soft security threats were the least likely (71%). *"No self-respecting pirate is going to operate in the Arctic, and [regarding] the transport of illegal goods, if you land illegal goods up in the Arctic, what the hell are you going to do with them?"*

As far as rankings were concerned, most individuals rated cruise ship incidents first (53%), followed by a smaller number of individuals who rated ship collisions as most likely (47%). Oil spills (41%), groundings (34%), and risks stemming from smaller ships operating in the arctic (12%) were less likely to be rated as the most likely incident to occur. However, levels of agreement were much higher when examining the top two most likely incidents, with 82% ranking a cruise ship disaster as either first or second, 71% ranking an oil spill as either first or second, and 65% ranking a ship collision as either first or second most likely.



The Polar Code

Another theme we tried to gather from our interviews were stakeholder's perceptions of the effectiveness of the polar code and where they could identify room for improvement. From the sample of interviewees we gathered, all but one had enough familiarity with the Polar Code to comment on its strengths and weaknesses. The group agreed that there was indeed a need for the Polar Code because of the unique nature of maritime activities in

polar waters but disagreed on where it needed to be improved or if it needed adjustments at all.

One theme that emerged was concerns about enforcement. Ten of the 20 respondents mentioned a gap in the enforcement of the polar code in its current form. In many cases, they brought up that even mandatory provisions of the polar code relied largely on operator claims of compliance. For example, the provision on requirements for five-day survival equipment was brought up by a high-level Coast Guard responder in Iceland, telling us that " *it is not possible to fulfill it. Research has shown that the personal life raft cannot… they will just survive for one or two days*." This may indicate a disconnect between those who are making such regulations and those who are being expected to comply with them.

Other recommendations for improvement included some that believed that the Polar Code did not do enough in terms of environmental regulation – particularly in the form of a heavy fuel oil ban and waste water dumping. Some respondents believed that the boundaries of the polar code themselves seemed to be incorrect. Noting that "*It's political of course and financial because it excludes areas of traffic [like Iceland."* The issue of boundaries was also brought up in the uniformity of regulations in the Polar Code's coverage. *"It was a little bit strange to put the same requirement in the whole area."* Greenland, for example, has the same mandates under the provisions of the Polar Code at its northern and southernmost points. On the other hand, the tendency of the requirements to treat the area uniquely or to have different requirements between states was brought up by two of our respondents, for the fear of business skirting those with more regulations to areas without.

Another common response we had to the question of improving the Polar Code from a few of our respondents was that the Polar Code is so new and that both industry and the international community required some time to see how well it works in its current state before any changes should be made. "*we need some time to both be sure that we understand the provisions and see that it is applied the same way all over [the region]."*

International Cooperation

We also surveyed respondents' perceptions of international cooperation in the region to get a sense of how well different states and their corresponding agencies and departments worked with each other. Every respondent described cooperation in the region as good. However, there were several caveats they discussed with similar themes emerging from their criticisms.

Common Goals

Seven respondents made sure to note that current levels of cooperation should not be taken for granted, or noted the need to be wary of what one respondent called "arctic exceptionalism" in the future. Cooperation, to these respondents, was happening in areas where state interests lined up in such a way that cooperation was the best way to achieve these interests. Citing as examples the Arctic Council's SAR and MOSPA agreements, one respondent noted that "*the issues that are cooperative are of course in everyone's interest to cooperate.*" There were, however, concerns raised by some of our respondents regarding international cooperation, mostly centering around Russia and China.

Uncommon Goals

There were a few respondents who believed that Russia would disrupt the current state of cooperation in the region. One respondent noted that "*Russia might not be as involved as we'd hope [in cooperative fora]*" and another stated that "*I challenge you to find anything after 2014 where any of the countries were willing to surrender interest to the other side*" – referring to Russia's annexation of Crimea and subsequent backlash form the international community perhaps causing a disturbance in the standard of cooperation in the region.

This view was not shared by all of our respondents. Others believed that geopolitical concerns may not have as big of an impact on the day to day operations in the Arctic. One person familiar with Coast Guard operations in areas patrolled by Canada, the United States and Russia told us that *"[First responders] are cooperating on search and rescue. You know, it doesn't matter what happens up there when somebody's life is in danger, whether you're a Russian coast guard, American Coast Guard, Canadian coast guard, you pull the stops and you go. I know that's happening today."*

Are Any Additional Agreements Needed?

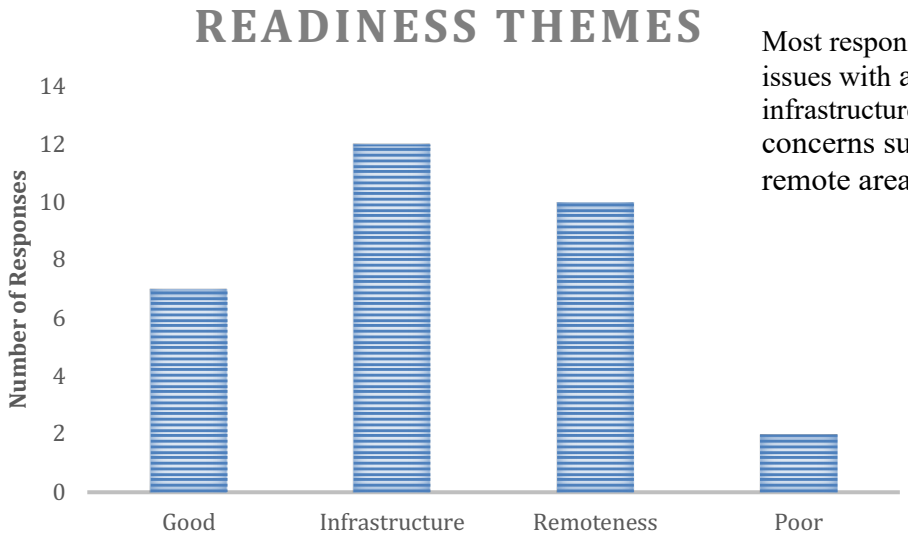
After getting a sense of the cooperation in the region, we asked our respondents if there were any other agreements that should be considered by international community. Of the 15 that answered the question, 11 told us that there currently appeared to be a good framework for international cooperation, but expressed a need to ensure that the ones that already existed were being implemented. One Canadian academic asked rhetorically, *"[I don't think] that somehow a new international agreement will magically solve all these problems. I think to a large degree, the issue is, 'okay, how do we implement, how do we actually cooperate under these terms?'"*– an opinion that was echoed by others both in Coast Guards and maritime industries.

Of those who believed we needed *more* international agreements, two believed that there should be more environmental preservation agreements in the banning of Heavy Fuel Oils or waste water dumping. One suggested a ban on certain weapons in the Arctic could be fruitful, while another brought up security interests that should be discussed more in the Arctic. Therefore, the sense from the interviews is that in terms of agreements, there are enough, but they need to be implemented.

Gaps in Readiness and Capabilities

After discussing some of the venues in which countries interact in terms of policy creation and other higher-level interactions, we asked our interviewees about some of the actual operational capabilities of search and rescue and environmental response. We received a wide range of responses on this subject in terms of perceptions of preparedness from "very good" to an unequivocal "poor". This spectrum of responses might be attributed to a particular region within the Arctic North Atlantic. This continues to illustrate its lack of uniformity across countries and their respective obligations for search and rescue and environmental response – especially as we increase latitude. Part of this discussion about the gaps in readiness came also from the physical resources that are

available to actually perform the operations by operators. Below is a visual of how respondents characterized readiness.



Most respondents noted issues with assets or infrastructure, followed by concerns surrounding remote areas

Infrastructure

Of the 14 respondents who answered our question about readiness to respond to a search and rescue or environmental disaster, 12 mentioned some lack of infrastructure in the region. Our respondents described infrastructure as things from terrestrial resources like airports, communication bandwidth, and roads to sea and air-based assets like ports, ships, both ice-capable and otherwise, helicopters and equipment for responding operators.

Our respondents also noted how a lack of infrastructure could affect sparsely-populated areas and their communities. When considering how to respond to a disaster, either lifesaving or environmental response, one respondent noted the importance of having sufficient resources, "*even if it's temporary and we know that we can go and set up [in] a small community in a really harsh environment and survive, and not impact the local community and use all their resources*." Our respondents noted that even basic survival needs like food and water, shelter and restroom facilities would be in short supply for both the disaster responders and those who would need to be brought ashore in the case of a mass casualty event. When considering cruise ships that have as many as 3000 passengers arriving in villages with sufficient resources to sustain the lives of 1500, it's easy to imagine a situation that quickly becomes unmanageable, even with the best coordination and planning.

Remoteness

Another common response we received in response to our question about readiness in the region had to do with the vast distances and harsh conditions that are covered by the relatively small numbers of assets in the area. One respondent, who works for a maritime insurance agency in Greenland stated that "*Joint Arctic Command has one helicopter they can send out in case of such a rescue operation in Greenland, the area covers what is the size of India.*" Concerns about sufficiency of resources in remote areas were

echoed by a Canadian respondent who warned that "*the farther north you go, the less you have infrastructure, the more problematic it becomes.*" The sense from these interviews is that there is good cooperation and coordination from rescue services, but in the event of a large-scale incident, no one denies that the community is ill-prepared.

"Good"

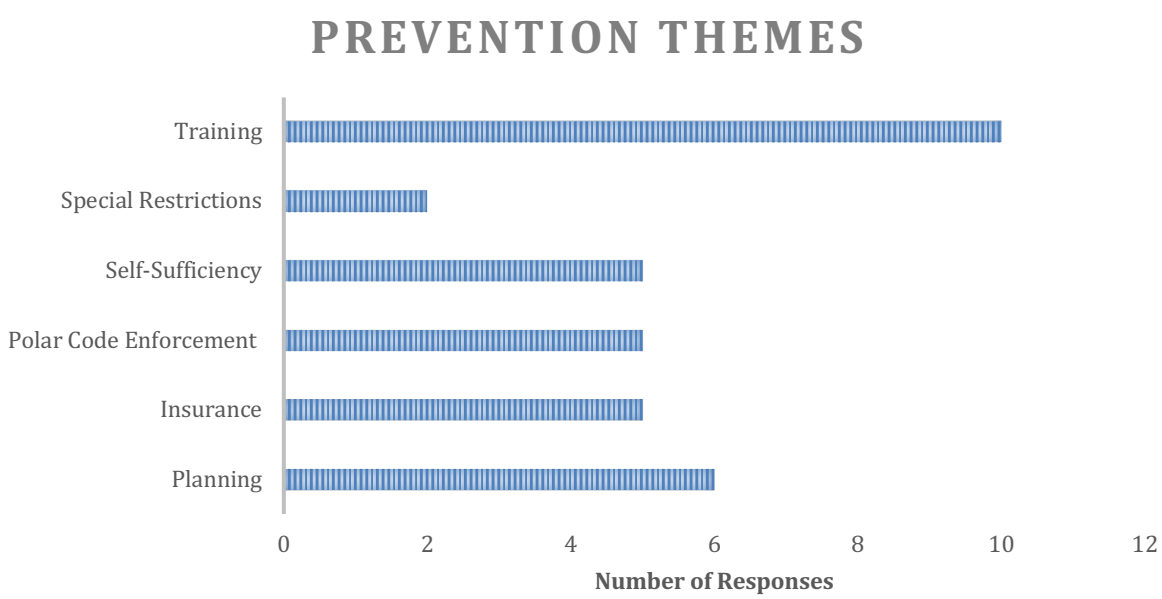
When questioned about whether or not increases should be made in terms of investments towards infrastructure in the region, we received a lot of reasons why this may not be feasible. Many told us that it came down to a basic risk assessment.

A Canadian industry respondent told us that "*you just have to step back and look at the bigger picture that being you know, like a percentage of overall Canadian vessel traffic that is happening in the Arctic, 99% of it is South so it doesn't make much sense from financial point of view, if I can be so blunt, to allocate a dedicated search and rescue station.*" Two Danish search and rescue operators also echoed this sentiment, pointing out that "*the question is, well you can do something about it, but it has a cost*" and that *"[if] a huge cruise ship capsizes it is not sufficient, but it wouldn't be sufficient anywhere in the world*." It would seem according to these emergency response operators, that preparedness is adequate for the relative risk in the region.

Others believed we should "*take [preparedness] seriously, but we need to get past the hyperbole [of the worst-case scenario]*." On the other hand, there were also those felt that we're already behind in needed investments. "*The problem is that you don't just snap your fingers. we need to be thinking about the fact that development is just going to happen. you don't just snap your fingers and get arctic assets. One thing that I guess I was fearful of is that by the time we realize woah, we're out of step here it's going to be further behind that we are now*."

Prevention of Arctic Incidents

Turning from reaction to prevention, we then asked our respondents their opinion on the best way to prevent accidents from occurring in the first place. We received several answers, ranging from improved weather and ice modeling for the region, to the role of insurance agencies, but most respondents indicated that the best way to prevent a disaster is to train for one.



Training and Exercising
Training and exercises, as mentioned here referred to cooperative exercises between states. Some of our respondents were familiar with the joint exercises that have been done in the past and recognized their value as a tool to identify gaps in cooperative response capabilities (n=10).

However, one respondent noted a need to expand beyond regional cooperation because "*we should [conduct] exercises with the nations that we never have to cooperate with.*" This sentiment was shared by another search and rescue operator who told us "*I think they're still small scale at least from what I've been able to see on the US side, because we just don't have a lot of the capabilities to do a lot of the things we say we want to do in the Arctic*." Thus, while respondents recognized the value of practical exercises, they also warned of some of the political motives associated with them.

Insurance
We specifically asked our respondents about their perceptions of the role that the insurance industry should play in preventing disasters. 19 respondents said that insurance should play some role in prevention, most without providing specific suggestions for how that might occur. A few of our respondents did give us some specific insight into the role the industry has played in the past and gave us an idea how it might fit into prevention in the future. They also specified where they are unlikely to intervene.

For example, our respondents noted that the insurance industry was proactive with regard to compliance with polar certificates. Another respondent who works in the maritime insurance industry specified her view of the insurance industry's role in making policy in the future. "we are a business partner and we are making a business together with the shipowner.... if they follow the rules that the international bodies have laid out, it's very difficult to ask for something more." Noting there that insurance companies will not take

the lead in creating policy, but would follow what national or international bodies require.

Importance of Planning
Another common response we received in terms of preventing incidents in the Arctic North Atlantic region was better planning through the improving of weather and ice forecasts as well as nautical charting. Weather forecasts appeared in six respondents' recommendations for preventing disasters in the more northern areas of the Arctic North Atlantic. Understanding that weather in this region can shift rapidly was a concern for many of those who study the region and those charged with responding to a disaster.

A common misconception that many of our respondents mentioned had to do with the notion of an "ice free arctic". One Canadian Coast Guard official expressed his frustration about reports of "ice free summers" and how they can mislead those who would expect different conditions than actually exist, because they do not account for floating icebergs which can cause catastrophic damage to mariners. Six respondents claimed that tracking these ice floes are important in preventing incidents in the region. A respondent familiar with ice mapping explained that while this is being done in other regions of the world, it has yet to be expanded to the regions we studied.

Similarly, nautical charting was also brought up by six of our respondents as a deficiency that leads to incidents. One participant also noted the lack of consistency across individual states' agencies charting as a cause for concern.

Self-Sufficiency
Five respondents highlighted the importance of private vessels to have the means to sustain themselves for long periods of time in the event of an incident, "*being able to get to safe harbor and having some ability to tie into whatever infrastructure is there [is important].*" Our participants mentioned a number of forms which self-SAR could occur, including detailed voyage planning, convoy travel and other means of rescue outside of coast guards.

Special Restrictions
Two respondents mentioned the possibility of applying special restrictions as a means of prevention. Examples included designating areas where ships should not be able to go, or regulations specifying that ships carrying a type of environmental hazard must stay away from an ecologically sensitive areas.

Are there Any Additional Threats?
We also asked our respondents if there were any additional threats that we had not discussed throughout the course of our conversations, or that were not being discussed by Arctic stakeholders. There were two recommendations that came up consistently, the introduction of new operators in the region and hard security threats.

New Arctic Operators
Five respondents described concerns about new operators that will come into the Arctic North Atlantic region as opportunities arise to explore new economic enterprises. Their concerns included both issues with governance, but also concerns at the operational level.

An American academic warned of "*this whole issue of domain awareness*" and an Icelandic Coast Guard operator told us that "*I think our main goal now is to educate people and inform,"* especially those who have crews and companies based out of non-Arctic states. These new Arctic operators tended to fit into people's perception of how industry will increase, and the risk associated with crews primarily used to operating in warmer climates, as among their biggest fears.

Hard Security Threats

While Arctic Security was not the focus of our research, five respondents brought up the threat of a military event in the region. The fact that it was brought up not only by our military interview subjects, but by experts in several fields, made it compelling enough to include in our discussion.

These respondents mentioned possible terrorist incidents, cyber security breaches, or a military confrontation originating in the Arctic. While few respondents elaborated on a scenario, one was able to articulate what many had alluded to, *"... [an increased] military presence and communication [failure could] escalate a very minor incident.*" These hard security threats, to these respondents, are among the issues that should be taken seriously in the region.

Forecasting the Future

The purpose of the current study was to prepare a foresight analysis for shipping activity in the Arctic and North Atlantic. Although not expressly intended to do so, our report also serves as an update on conditions noted in the 2009 AMSA report. Regrettably, we cannot conclude that there have been drastic improvements on the state of affairs since the time of the AMSA report, although progress has been made in some key areas. Even now, some ten years after the publication of the report, were we to make a list of concerns for the region, it would look very similar to the concerns noted in 2009. The good news, however, is that shipping and activity levels have not adhered to what was predicted; and up until now, major catastrophes have been avoided. In what could properly be termed the ambivalence of the Arctic, the fact that there have been so few major incidents can be seen as both a blessing and a curse, insomuch as many have noted that change in the Arctic is often reactive rather than proactive. Instead there is the dual sentiment that it is "just a matter of time" before something happens, and it is only then that major regulatory changes will occur.

Arctic Shipping is all about the Economics

The slow rate of change in the Arctic at present is being driven primarily by economic constraints that make increased trans-Arctic shipping more of a dream than a reality. As one of our interviewees from the shipping industry stated, "*the economics simply aren't there*." According to both qualitative and quantitative data examined in this study, the economics will likely remain unfavorable to large scale containerized shipping even into the near future, given the low price of oil, and unpredictability due to ice and weather inherent in shipping in the Arctic. Barring significant changes in the current geopolitical environment, or drastic increases in the price of oil, shipping in the Arctic is likely to remain largely restricted to destinational traffic which is predominant in the region. We take this to be a positive development, given limitations in the number of assets and infrastructure noted by so many of our interviewees.

We Might Need to Expand Our Focus

The failure of the shipping boon to materialize at present may also be an advantage given insurers' disinclination to step into a regulatory enforcement role in the Arctic. While insurers see themselves as part of the solution for ensuring safety in Arctic shipping[68], they remain unwilling to increasing the cost of doing business for risk prone ships. Our insurers, similar to those surveyed in prior research, reiterated their unwillingness to serve in a regulatory or enforcement capacity, stating that it runs counter to their business model of partnering with shipping companies. When pressed about what would happen if a catastrophic incident occurred, insurers noted that what would change were the regulations, not the role that insurers play, preferring a more reactive, rather than proactive, stance.

One of the most striking things that we came across in our research is the lack of concordance between what the data shows regarding the most likely incidents in the Arctic, and those that were mentioned by the interview participants. On the whole, the issues that concerned our interview respondents were simply not issues that were the

most likely to occur from a historical perspective. This suggests a need to separate out the 'day-to-day' operations in the Arctic from the 'worst-case scenario' that may occur.

Thankfully, the worst-case scenarios remain among the least likely, but with an increase in both the number and size of intra-Arctic ships, there has been a corresponding increase in incidents. Whereas some have said that the risk of an accident happening in the Arctic is the lowest it has ever been in terms of ship losses,[69] our analyses were able to show that the number of smaller scale incidents is steadily increasing. Although the data accessible for the present study precluded the ability to determine the exact size of the correlation between the increase in traffic and the numbers of shipping incidents, it is not hard to imagine that these two are likely related.

And yet, not all increases that are occurring in the area present equal cause for concern. For example, our interview participants seemed less concerned about the expansion of veteran, more established industrial operations. It was when increases exceeded the expertise threshold, when expansion was accompanied by more unprepared or inexperienced individuals operating in the region, that our interview participants felt that increased shipping posed a problem.

Many indicators, however suggest that an increase in novice operators may be on the horizon, stemming from industries that have the current economic motivation and ability to expand. Both cruise tourism and fishing are predicted to show expansive growth in the coming years, and both serve as industries most likely to bring inexperienced operators to the region. Combined with the fact that a large-scale cruise ship accident is everyone's worst-case scenario, and it is easy to see why the time for concern may be sooner rather than later. Especially because, according to our respondents, we simply do not have sufficient assets and infrastructure to deal with a large number of passengers in an emergency situation in the Arctic.

Coincidentally, while this project was in press, a cruise ship incident occurred involving the *Viking Sky* off the coast of Norway. Boasting of being the first U.S. cruise ship to offer a full-length itinerary in the Arctic Circle during the winter season[70], this cruise ship carrying almost 1400 passengers and crew nearly ran aground in rough weather. The incident resulted in the rescue of around 470 passengers over the course of 20 hours, utilizing two rescue helicopters able to carry only 20 people at a time. Twenty of those rescued required hospitalization.[71] If the ship had been in a more remote location, this could have easily been that worst-case scenario on everyone's minds. In fact, one of our interviewees followed up with us after the event stating*, "Imagine[the] same off the coast of Northern Canada or Greenland – and you have your catastrophic incident…."*

The story of the *Viking Sky* is not unique compared to historical accounts of rescues in Arctic shipping. Despite the high visibility of the dangers of the incident, the *Viking Sky* also speaks to the importance of a well-trained crew with self-sustainable rescue capabilities. What this and many other incidents share is a story of rescue operations that could have been so much worse, given the current state of SAR readiness and capacity. While nearly all accounts concur that resources are suboptimal at present levels, there have been relatively few human casualties stemming from marine incidents in the Arctic. The number of human casualties from cruise ship incidents is even lower, according to

the sample of data we have (around 0.2%). But with planned increases in cruise ship activity, and an additional 30 ships on order for the purposes of expedition style cruising, the likelihood of an incident occurring far from rescue resources will also increase.

The other industry experiencing rapid growth in the region, presenting a heightened potential for accidents, is fishing. Fishing vessels are twice as likely to be involved in a marine incident than cargo ships, and are almost three times as likely to have repeated casualties. These statistics are concerning for a number of reasons, including their overrepresentation in the Arctic shipping fleet, but also in light of their own economic incentives to follow fish stocks to more remote areas. The fact that fishing vessels are not currently subject to the regulations of the Polar Code was mentioned as a concern by a number of our experts. Therefore, both the qualitative and quantitative data in our study suggests that it may be time to revisit extending the requirements of the Polar Code to fishing vessels. Timing may now be of the essence, before the fishing moratorium runs out and there are even more fishing vessels plying Arctic waters, and before even more stakeholders may be motivated to fight an expansion in regulations.

Thus, as is clear from our analyses, there exists a disconnect at present with regard to disasters in the Arctic. The disaster everyone fears, which has a lower likelihood of occurrence, and the disaster that is most common, but is not as prominent in everyone's minds. Future work in the area should continue to monitor and prepare for both types of disasters if we are to effectively manage the safety of maritime operations in the ANA region.

Joint Exercise Gaps

Given the tendency for Arctic stakeholders to focus on the "hyperbole of the worst-case scenario," it is perhaps unsurprising that tabletop exercises, currently the primary means of preparing for disasters in the Arctic, also show this predisposition. Six out of 11 live or notional exercises have focused on a cruise ship disaster, with 5 involving preparation for all other types of marine incidents combined. The timing of these incidents may also be a factor, in that our analyses indicate that the most common months for accidents in the Arctic are in February and August (with August being the most common month for disasters in the ANA region). The fact that most live exercises are conducted in the summer months, likely due to constraints of availability, again suggests limitations in our preparatory and readiness capabilities. The recent redesign of Operation Nanook to include exercises in March, and expanded to including a larger number of representatives from Arctic states, is a step in the right direction, and it is our hope that other exercises will follow suit.[72]

We also recommend, where possible an expansion of exercises to include industry and local community participation as well. Given the lack of assets and infrastructure, it becomes even more important to include the emergency response assets of both of these stakeholders in SAR and environmental operations in the area.

Finally, there is a tendency for exercise scenarios to focus on either rescue or oil spill intervention. The number of incidents involving both in the incident database points to the need for more expanded and inclusive preparations to enhance our readiness and

capabilities to respond to the accidents that are happening, in addition to the ones we most fear.

Vessels of Opportunity

Additionally, when we think of search and rescue assets, there is a tendency to consider only emergency responder assets and vessels, to the exclusion of the most common asset in actual SAR incidents. Historically, in the ANA region, as in the Arctic as a whole, the most common vessel responding in rescue operations is vessels of opportunity. This is perhaps unsurprising given the vast distances and lack of permanent infrastructure in the area. The fact remains that the relative safety of the Arctic has occurred in large part due to the presence of vessels of opportunity, and expansion of preparations that include coordination of these vessels would be beneficial. We echo the sentiment expressed in the after action reports, and recommend that industry should be included in emergency preparations whenever possible, as these assets are likely to be needed in to respond to incidents in the Arctic.

In another example of the ambivalence of the Arctic, while cruise ships and fishing vessels pose the greatest threat, they also represent the most promising vessels of opportunity. Should a disaster happen, cruise ships are more likely to have assets on board that would be needed in times of crisis (e.g., lifeboats, medical staff and resources, food, space, etc.); and, historically, fishing vessels have been the most common SAR asset after lifeboats and helicopters.

Given the harsh conditions of the Arctic, however, vessels of opportunity may fall victim to similar problems, and become ships in distress themselves. As seen with the recent incident involving the *Viking Sky*, the cargo ship diverted to assist also suffered engine failure, and rescue helicopters had to be diverted away from the cruise ship in order to rescue the crew of the cargo ship.

In any event, it is clear that rescue preparations should continue, and insomuch as they are able, should include both worst-case scenario preparation, as well as preparation for the more common scenarios which may be encountered in the Arctic. The coordination and use of commercial vessels of opportunity in emergency response scenarios should be expanded, as they are the most likely rescue asset utilized in incidents so far.

The Future of the Polar Code

Expansion and clarity were also among the many suggestions for the future of the Polar Code. Perhaps unsurprisingly, responses from industry representatives tended to adopt a "wait and see" approach, while responses from SAR responders tended to call for more stringency and specificity to the provisions. Despite the lack of consensus as to the future of the Polar Code, one area in which the data and interviews agree related to the 5 day survivability requirement, as specified in the Polar Code. To date, no study has confirmed survivability in a lifeboat for that length of time. Combined with the fact that lifeboats and liferafts were among the most frequently utilized rescue assets, and the need for additional research and development is obvious.

Representatives from all industries were also likely to call for greater specificity and more stringent requirements in the area of crew training. There was a near universal

concern over the possibility of new inexperienced crew coming into an area where experience is crucial. Therefore, strengthening the training requirements for all crew (not just officers), and requiring actual ice navigation experience, are advisable based on the increasing levels of incidents over the years, and the expected increases in traffic to come.

Finally, we would recommend that the mandatory provisions of the Polar Code be expanded to include the areas which are most incident prone. This would include Murmansk, and parts of coastal Norway. According to one of our interview participants, *"[The Polar Code affects] areas with very little traffic anyway. So the areas that really need heavy regulations are excluded."* Data on incident rates in the ANA region confirm that these areas are among the most incident prone, so ensuring that safety regulations extend to these areas is important.

Expanding Safety Utilizing Existing Resources

Calls for increased investment into new assets and infrastructure such as icebreakers, helicopters and oil spill response capacity are not new, and thus need not be repeated here. Until such time as capacity increases to meet the level of risk in the Arctic, it is clear that such investments should be made. However it is our belief that targeted investments, aimed at strengthening existing resources, may also greatly improve Arctic marine safety. The following recommendations involve capitalizing on the fact that vessels of opportunity are the most likely first responders to incidents in the Arctic.

Training of Additional First Responders. Because professional first responders from state agencies may be days away when an incident occurs in the Arctic, training crew of all vessels to assist in smaller scale incidents in the Arctic would improve emergency response capacity. Similar to recent calls for local communities to be trained to assist in marine incidents,[73] it seems advisable that crew members operating vessels in the area might also receive such training, especially as they become more likely to be called on to respond when something goes wrong. Additional training to include oil spill response capabilities, for crewmembers to mitigate problems until professional help arrives may also be advisable.

Expansion of Rescue Resources. Again capitalizing on the fact that commercial vessels are likely to be closer in times of crisis than first responders might be, we also propose a policy that expands the SAR resources that are required to be on board vessels operating in the Arctic. Requiring or incentivizing ship operators through subsidies, to provision more than just sufficient resources for self-SAR would be a cost-effective way to ensure that rescue resources are available in the areas in which they are most needed. Providing incentives for private companies to secure additional rescue equipment, oil spill intervention equipment, or to carry more equipment than necessary would allow for an expansion of rescue capacity at a low cost. This would mean that Coast Guard resources which may be days away, may only be hours away on board another ship, and policies examining how to encourage private purchase and provisioning of rescue resources may bear further examination.

Expansion of Shipping Databases and Open Access to Information. The above mentioned initiatives, however, are much less useful if no one knows which ships are in the area, or what capabilities and resources they have. One suggestion to alleviate this problem is by combining two existing regulations that currently apply to ships sailing in Arctic waters. First, participation in Long Range Information Tracking (LRIT) is mandated under SOLAS, for all passenger and cargo ships over 300 gross tons engaged in international voyages, to provide regular updates as to their location.[74] Additionally, the Polar Code requires ships to prepare a Polar Water Operational Manual, which includes ship specific capabilities and limitations with regard to SAR procedures, and plans for ensuring safe operations and life support, if needed. Combining these two requirements, we propose expansion of the LRIT database to include a mandatory filing of all ships' Polar Water Operational Manuals, and we propose that this database be accessible to all relevant rescue responders. Expanding information required on file through LRIT would help rescue coordinators make more informed decisions regarding the resources that may exist in the area. This in turn will mean more effective deployment of the most needed resources, especially if a ship of opportunity were to be called upon to assist in an emergency. The Polar Code already requires preparation of this vessel plan for ships sailing in the Arctic, having it on file somewhere where rescue operators can access it seems like the next logical step.

Having access to information regarding a ship's position and capabilities would also be useful for more than just reacting to incidents as they occur. Having an open database for ship traffic and capabilities would also allow for more proactive research to be conducted regarding trends and best practices. The lack of data on ships, and ship incidents is astounding for an area that claims to be concerned with enhancing ship safety. The fact that this information is currently hidden behind paywalls means that relevant information is not accessible to those who may need it most. As it stands, Arctic stakeholders have very limited access to free information regarding ship incidents, and the information that is available is inconsistent or hard to combine. Even the Arctic Council's own PAME database, the ATSD requires a fee to access, making the development of Arctic marine best practices much more difficult. (So much so, that we were never able to get access to this database for use in the current study.)

Expanding Charting and Hydrography. Hull and machinery damage, and wrecked/stranded vessels were the two most commonly occurring incidents in the Arctic, according to historical data. It is not surprising then that a consistent gap identified by coast guard operators in our study was the need to update charting and maps, which may in some cases be over 30 years old, and in others nonexistent. Combined with the fact that melting ice will continue to open up areas that no one has navigated, means that more attention should be paid to charting, hydrography and navigational technology. Again, capitalizing on the potential offered by existing vessels operating in the Arctic, we encourage examination of the extent to which symbiotic relationships through public private partnerships can occur. Ships that are currently operating in the region could transmit data that would assist businesses in designing technology that industry needs. It would also make it much easier for ships to transmit data that is mandated (e.g., ship locations to LRIT). As noted earlier, making data easier to be collected and disseminated is in everyone's interest and will make the Arctic a safer place to do business.

Conclusions

Despite the fact that by all accounts, existing SAR resources do not match the level of risk in the Arctic, our study has shown that the Arctic remains a relatively safe place to do business (as compared to other parts of the world). However, just because the area has avoided a major catastrophic incident thus far, does not safeguard against the future. In other words, while the probability of a mass casualty or mass pollution event is low, the severity of either of these events would be immeasurable. There is a troubling consensus as to the fact that "it is just a matter of time" before the worst-case scenario happens, which everyone agrees will take the form of a large cruise ship disaster. With the recent near-grounding of the *Viking Sky,* it is clear that these fears are not without merit. But the truth remains that other types of incidents remain far more likely to occur in the ANA region than this worst-case disaster, and we should increase readiness and capabilities to respond to these types of disasters as well. Putting all of our attention and resources into SAR capabilities, to the exclusion of charting and hydrography, for example, may mean that we fail to address the much more common incidents that are happening now.

International cooperation is rated favorably in the area, and a mandatory regulatory framework has recently been put in place that will make shipping in the Arctic safer. Concerns regarding enforcement of those regulations, as well as the limits to international cooperation loom large in the minds of those who we interviewed. For these and many other reasons, it is perhaps a good thing that the predicted explosion in Arctic shipping has, at least for now, seemingly been placed on hold. But there are increases coming in industries that pose a great deal of threat to the area. With an additional 30 expedition style cruise ships on order, additional scrutiny is prudent to ensure that development happens in a responsible manner. Further, expanding the requirements of the Polar Code to fishing vessels, and doing it sooner rather than later, makes logical sense according to their safety record in the Arctic, and the impending end of the fishing moratorium.

Given the current economic climate, it seems unlikely that large scale investments in assets and infrastructure are on the horizon for any of the Arctic states (save Russia). This means that every effort to make the most of the resources that exist, bears consideration. This is happening now in the form of cooperative agreements and joint training exercises, but there are a number of other areas we think should be explored. Considering alternative ways we could position rescue resources in remote areas, without requiring large state investments in permanent infrastructure, could have a sizeable impact on marine safety. Capitalizing on commercial vessels of opportunity is advisable in light of the fact that the majority of incidents are smaller in scale, and are most likely responded to by vessels of opportunity first. Initiatives aimed at expanding upon local capacities should also be explored. In sum, by over focusing on catastrophic incidents, we may miss opportunities to make a real difference now, utilizing the resources we currently have.

A full list of our recommendations can be found below:

Recommendations:

1. Explore public private partnerships in emergency management
2. Explore potential inherent in vessels of opportunity
3. Establish unsecured channels of communication for information sharing among Arctic States
4. Enhanced data sharing and establish a repository of openly accessible data on ship incidents
    a. Expand Long-Range Information Tracking (LRIT) to include:
        i. Polar Water Operational Manual for all ships operating in Arctic waters
        ii. Rescue equipment and resources on board
        iii. Ship capacity
        iv. Medical or first aid resources on board
5. Expand training exercises
    a. to non-summer months
    b. to include more participation by industry and non-governmental groups
6. Increase coverage of Polar Code
    a. To include fishing vessels
    b. To include incident prone regions that are not currently covered by the Polar Code (e.g., Murmansk, Coastal Norway, Iceland)
7. Expand training requirements of the Polar Code to include actual ice experience, and expand to include all crew
8. Look for ways to engage expertise of local communities regarding safe zones, weather trends, etc. Capture existing "institutional knowledge" while exploring opportunities for strengthening ties to Indigenous groups, and providing real avenues for stakeholder inclusion.
9. Communities should consider adopting their own regulations and mechanisms of enforcement to suit their own regions and zones to capture the unique needs of each area while working to ensure that ships do not have too many disparate regulations to deal with as they cross areas.
10. Increased funding for research to examine the effects of oil spills in these areas, including continued investigation of the role of HFO in international shipping in general (not just Arctic).
11. Continue strengthening ties through existing organizations (e.g., Arctic Council ACGF and EPPR). Work to streamline operations so each has clear, minimally overlapping duties.

It should be noted that there is some overlap between our recommendations and those proposed in the AMSA (2009) report. This should not be surprising given that, as noted earlier, our list of concerns remain very similar to those raised in the original report. Our report shows that safety in the Arctic remains a work in progress, even some 10 years later. Considering alternative solutions to the recommendations which have already been proposed may prove fruitful in designing smaller scale interventions which can impact shipping safety, until such time as infrastructure, readiness and capabilities match the level of risk in the ANA region.

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Appendix A
List of Interview Questions

Project Description:
Graduate students from Brown University are working with the U.S. Coast Guard's Center for Arctic Study and Policy (CASP) on a foresight analysis to define and characterize potential seaborne disasters, catastrophic incidents, and security threats in the Arctic and North Atlantic (ANA) region. This initial research will tie into the broader objectives of the Arctic and North Atlantic Security and Emergency Preparedness Network "ARCSAR" project spearheaded by Joint Regional Control Center North-Norway. In addition to defining and characterizing historical incidents in this region, the research will also consider whether the combination of economic potential, environmental change, and increased interest and traffic in the ANA region has increased the risk of a maritime incident requiring cooperation amongst governments, industry, and security organizations. The ANA region's threats include, but are not limited to, the potential for armed conflict, terrorism, piracy, ship collisions or accidents, natural disasters, and pollution and other environmental impacts. The below questions are meant to develop data from the numerous stakeholders which will then be analyzed and presented to the U.S. Coast Guard for further contribution to ARCSAR.

With your permission we will be recording the interview so as to facilitate with transcription. Our goal will be to attempt to discern patterns in the data, so responses to the interview questions are expected to be reported in the aggregate. Recordings will be deleted as soon as they are transcribed. Should we wish to utilize any direct quotes from you, we would ask your permission beforehand.

Interview Questions:

1. Can you describe your (or your organization's) level of familiarity with the North Atlantic maritime region (defined as region between Eastern edge of Canada and Norway?)
2. What areas of industrial development do you currently see in this region? Will that development increase or decrease in the 15-20?
3. What potential areas of industry do you think are ripe for development in the next 15-20 years?
4. How familiar are you with the 2009 Arctic Marine Shipping Assessment[1]?
5. Have you noticed any changes in the industry or area that have occurred since the 2009 Arctic Marine Shipping Assessment?
6. What would you consider to be a significant increase in shipping traffic in the Arctic region? What are the critical gaps or major points of failure in the current governance structure?
7. We are focusing on three main areas of threat in the region - seaborne disasters (such as ship collision, grounding), catastrophic incidents (including oil spill, cruise ship accident) and security threats (piracy, transport of illegal goods, human trafficking, etc.)- of these, which do you see as the greatest likelihood? Why?

8. Are you familiar with the provisions of the Polar Code? As the current international law protecting polar waters, how could we strengthen the polar code?
9. What is your perception of the level of international cooperation in this region? By cooperation, we mean information sharing, joint exercises, etc…
10. Is there a model of cooperation that would work better in this region? What? Why?
11. How would you characterize the readiness and capability (including training, assets and infrastructure) for large-scale search and rescue operations? Environmental response operations?
12. What additional agreements should be considered? Bilateral, multilateral, or regional?
13. Turning to prevention, what is the most effective means of planning for an eventual disaster in the North Atlantic? Insurance? Self-SAR? Polar Code Enforcement?
14. In your experience, are there any threats we have not discussed?
15. Finally, do you have any recommendations for people we should talk with or sources for data we should consult?

Dissemination of this report will be dependent on clearance of the U.S. Coast Guard.

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